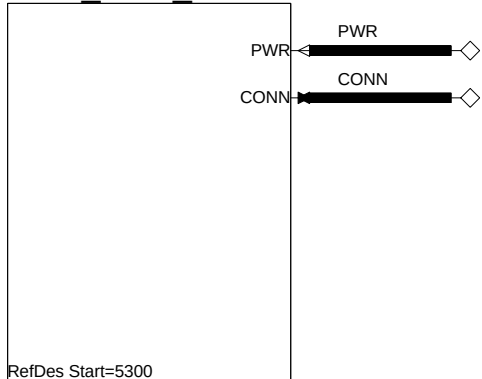


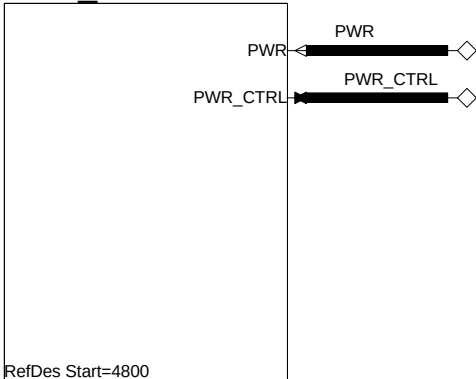
4010_Main_Connector



RefDes Start=5300

5300-5399

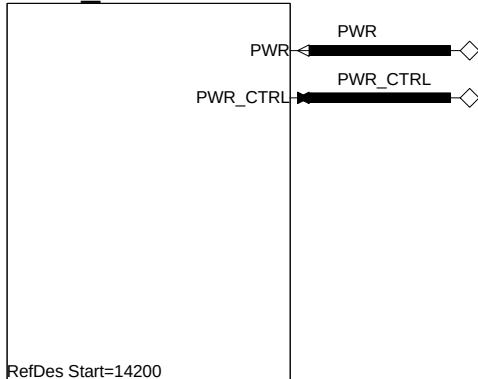
4010_Power



RefDes Start=4800

4800 - 4899

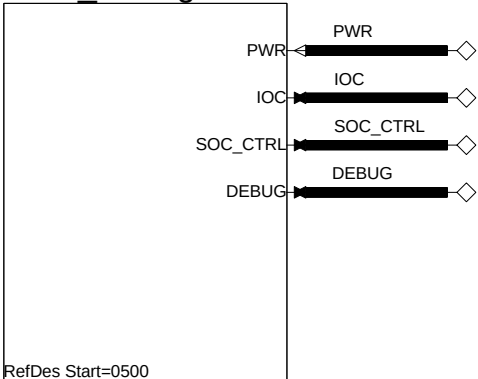
4010_PMIC1



RefDes Start=14200

14200 - 14299

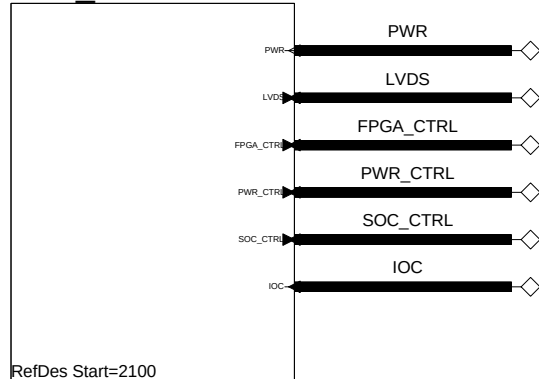
4010_Debug



RefDes Start=0500

0500 - 0599

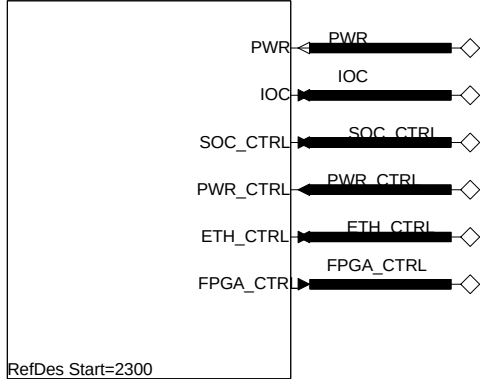
4010_FPGA



RefDes Start=2100

2100 - 2199

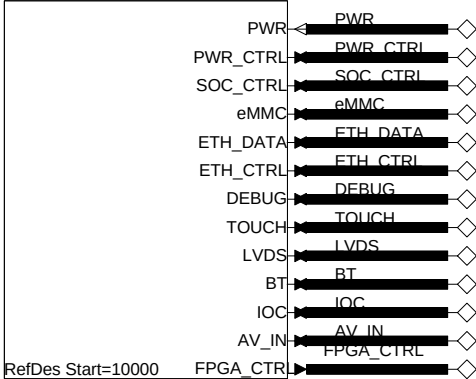
4010_IOC



RefDes Start=2300

2300 - 2399

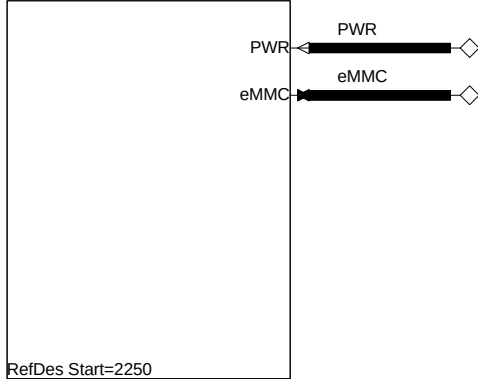
4010_SoC



RefDes Start=10000

10000 - 10899

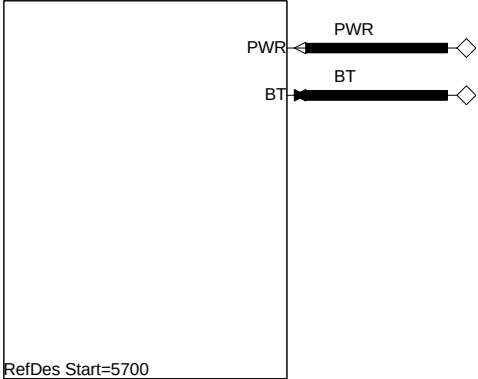
4010_eMMC



RefDes Start=2250

2250 - 2299

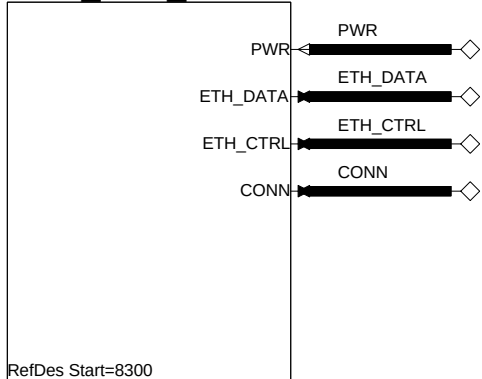
4010_Bluetooth



RefDes Start=5700

5700 - 5799

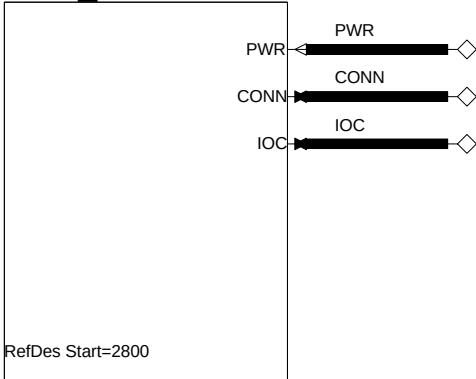
4010_ETH_PHY



RefDes Start=8300

8300 - 8399

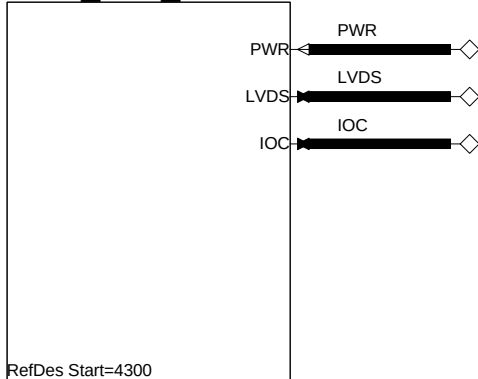
4010_CAN



RefDes Start=2800

2800 - 2899

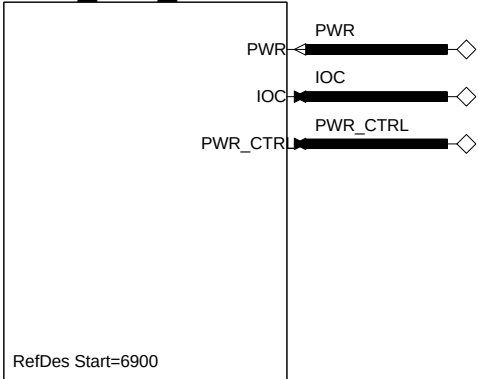
4010_TFT_Conn



RefDes Start=4300

4300 - 4349

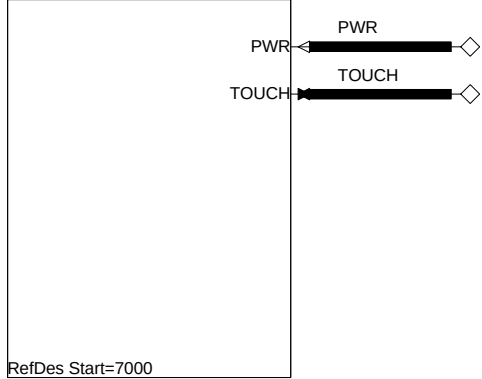
4010_TFT_BL



RefDes Start=6900

6900 - 6999

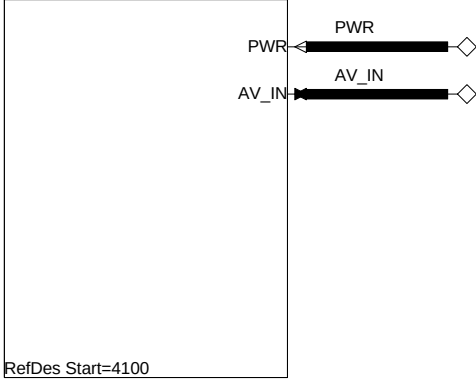
4010_CTP



RefDes Start=7000

7000 - 7099

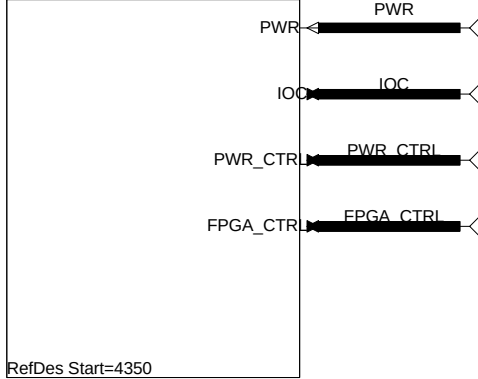
4010_VIDEO_DES



RefDes Start=4100

4100 - 4199

4010_TFT_BL_Conn



RefDes Start=4350

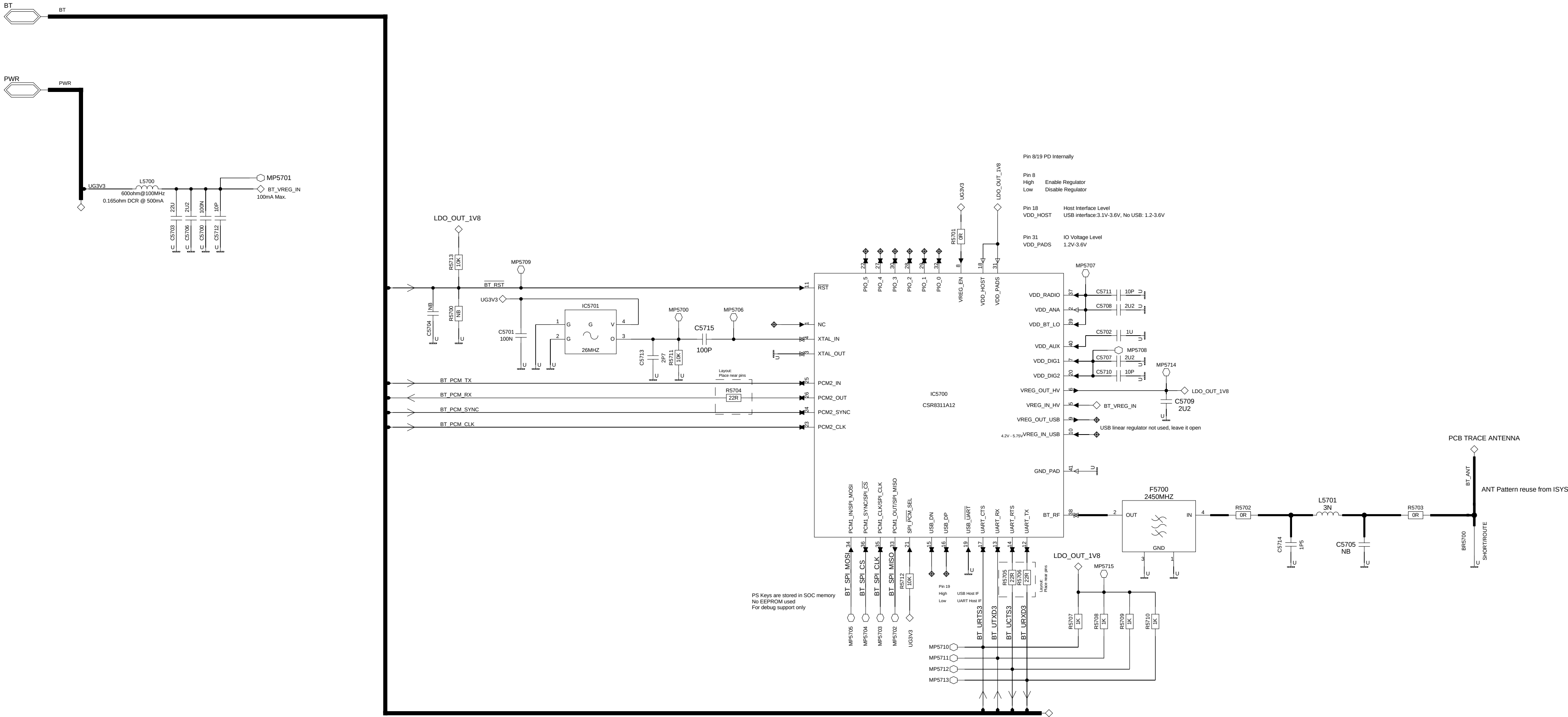
4350 - 4399

BC1
BARCODE-LABEL65X65
BC2
BARCODE-LABEL65X65

A	DV Change of topology for the backlight power supply	CN0008078	11-29-2023	FY	CZ
A	ED initial Release	CN0001912	07-30-2023	FY	CZ
Rev.	type of change	change-no.	date of change	auth.	checked
		Copyright reserved		Writing modifications reserved	
		created	11-29-2023	FYañez	
		checked	11-29-2023	CZamora	
schematic	SCHEM 4010 MAIN C143		Sheet	Sample	
Root_4010	FCA FRONT PASSENGER DISPLAY MODULE		1	31	DV1
substitute for:		Type		Part Number	
		C143		4106088	
substitute by:		Variant			
		C143			

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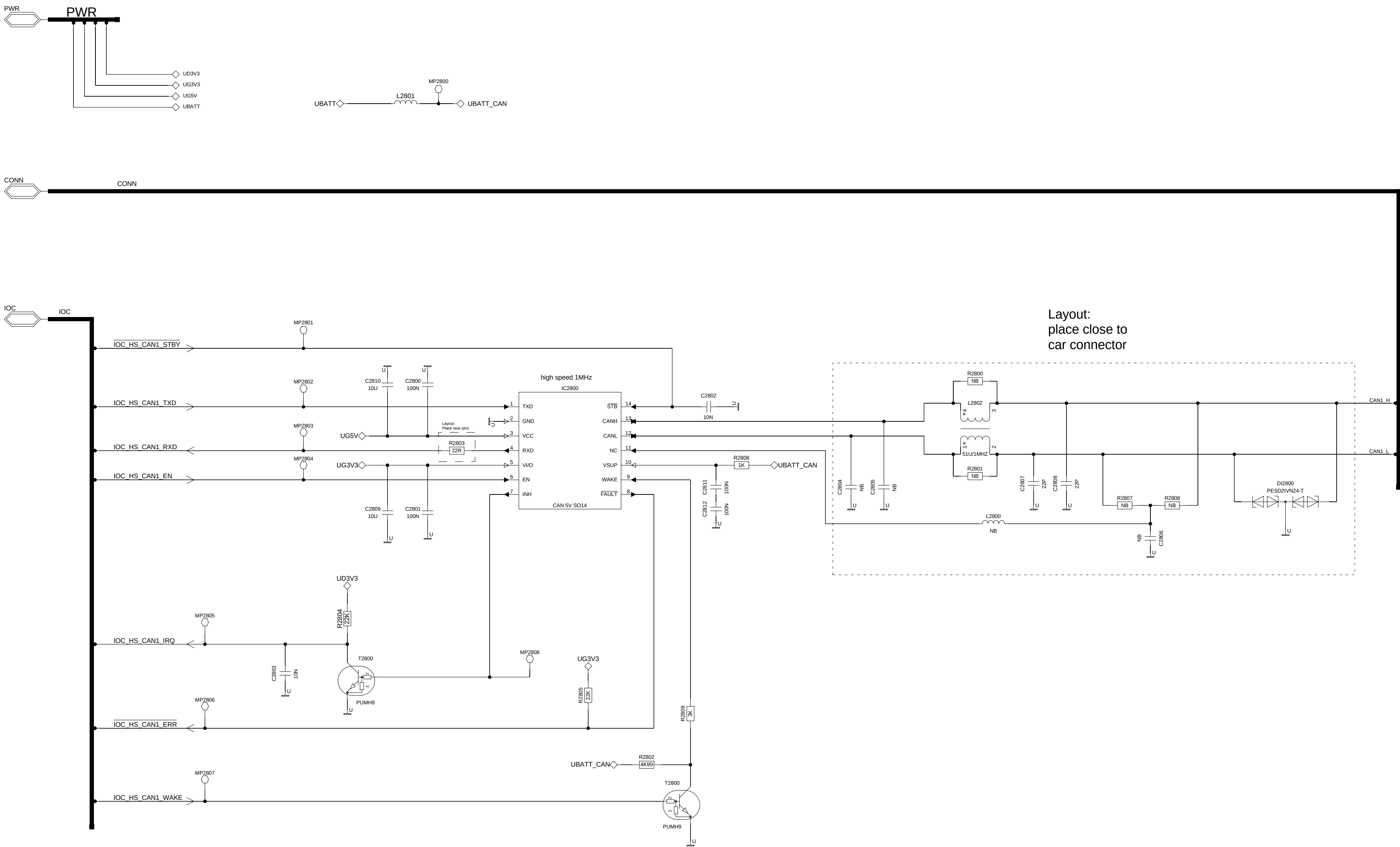
BLUETOOTH (5700-5799)



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Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	date		name	
		created			
		checked			
schematic	Sheets of		Sample		
4010_Bluetooth	2		31		
substitute for:	Type		Variant	Part Number	
		C143			

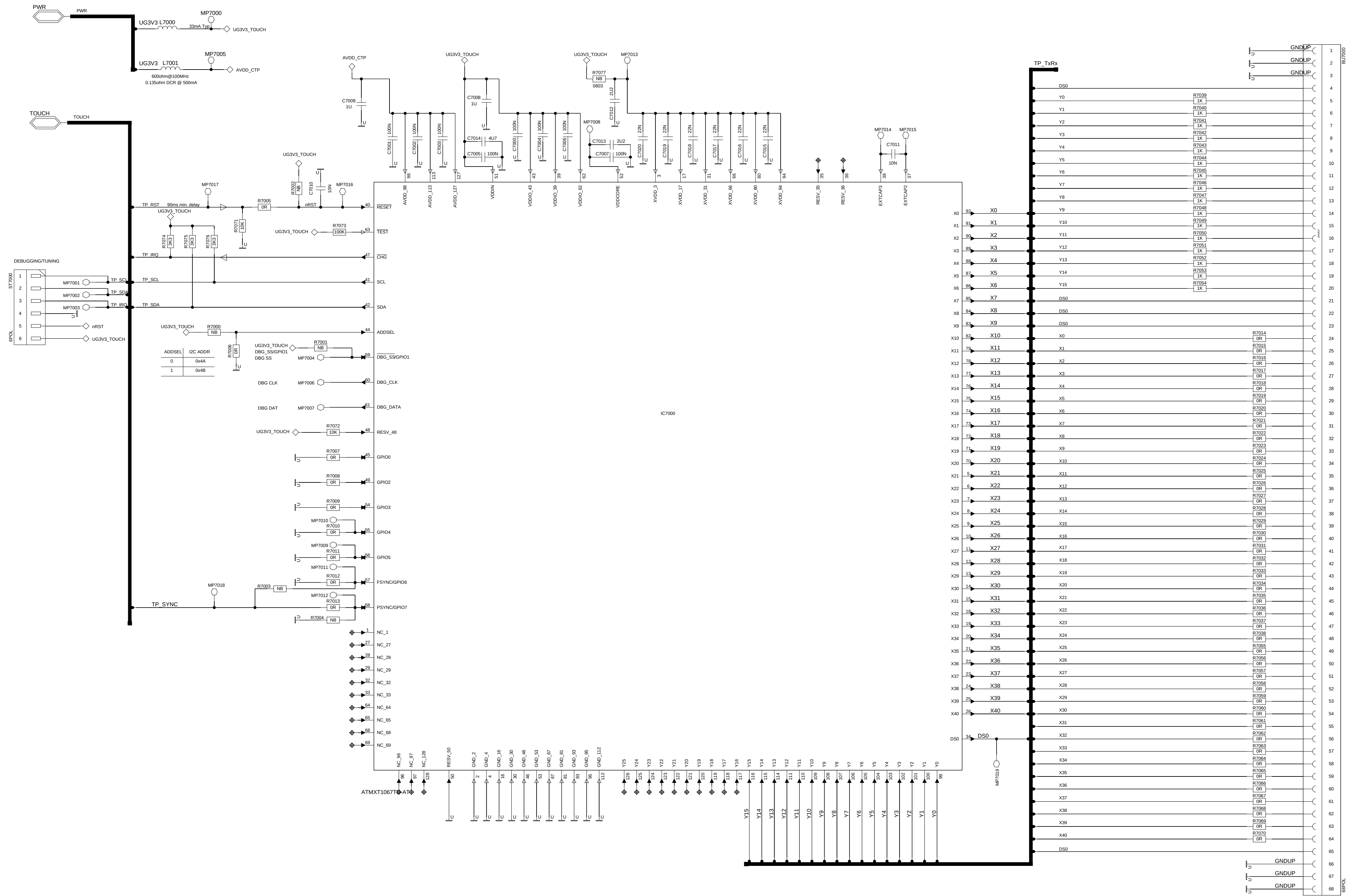
CAN (2800-2899)



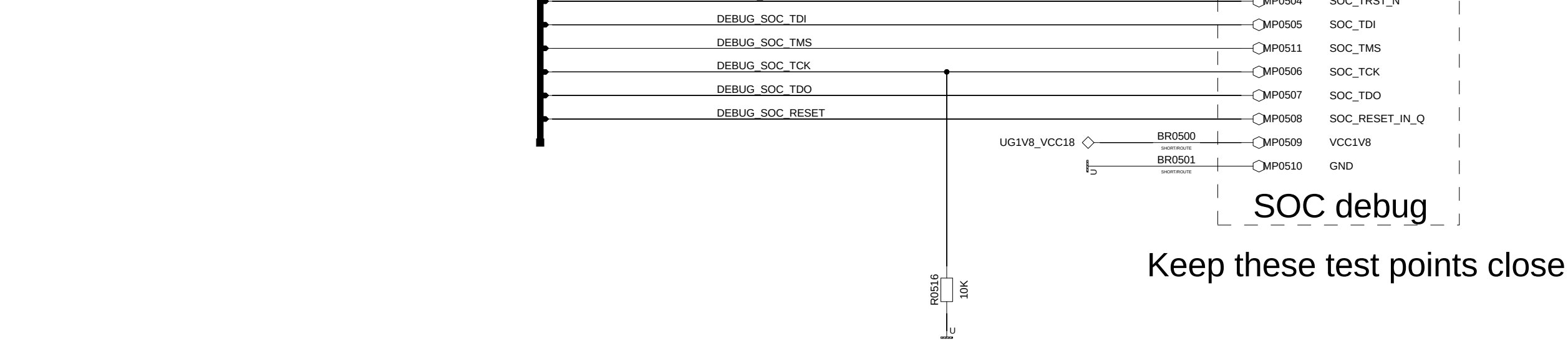
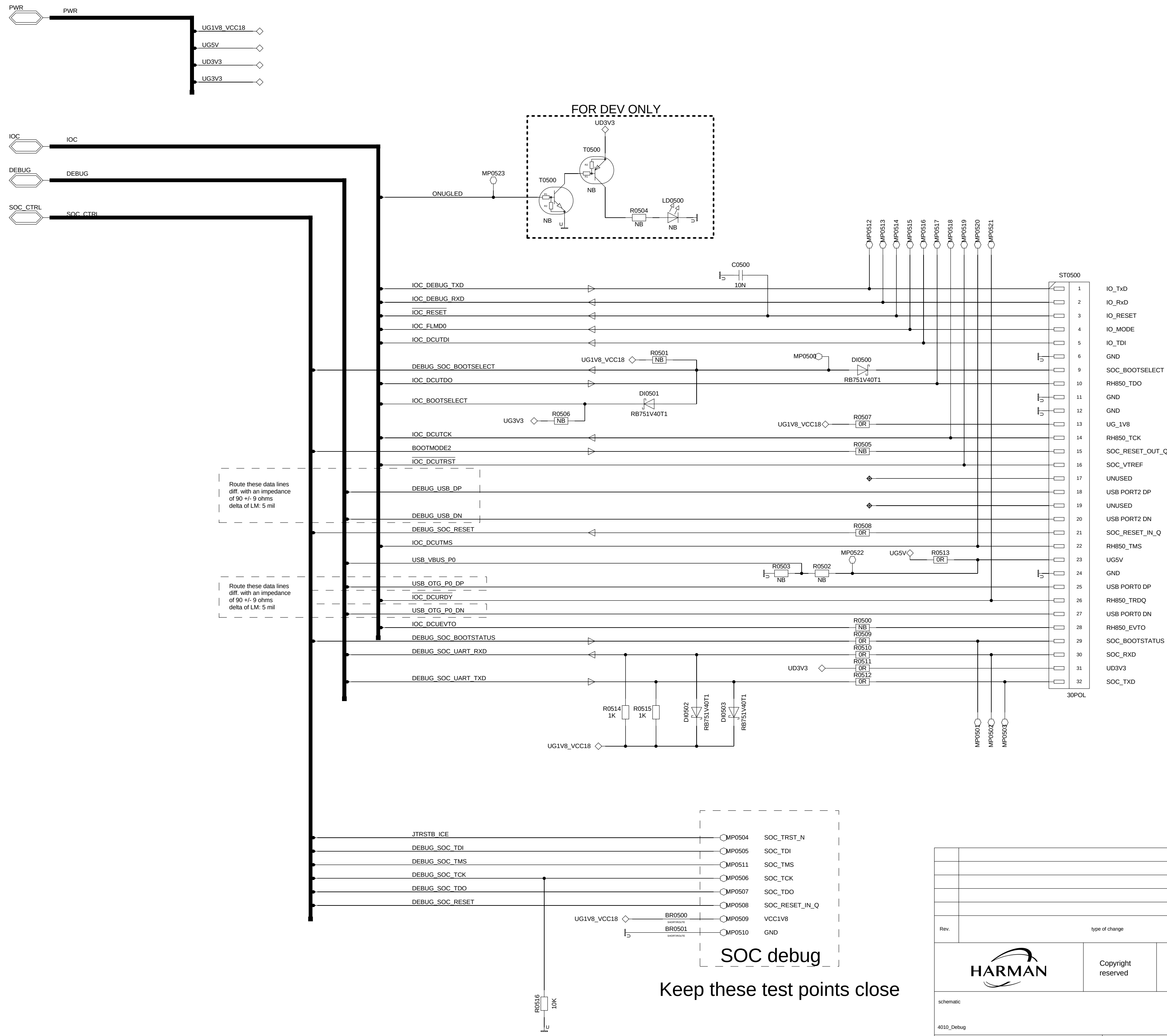
Layout:
place close to
car connector

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	date		name
			created		
			checked		
schematic			Sheet	Sheets of	Sample
4010_CAN			3	31	
substitute for:		substitute by:	Type	Variant	Part Number

CTP (7000-7099)



Rev.	type of change					change no.	date of change		auth.	checked
 Copyright reserved Writing modifications reserved						date		name		
						created				
						checked				
schematic						Sheet 4	Sheets of 31	Sample		
4010_CTP						Type	Variant C143		Part Number	
substitute for:						substitute by:				



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 5	Sheets of 31	Sample
4010_Debug			Type	Variant	Part Number
substitute for:		substitute by:			

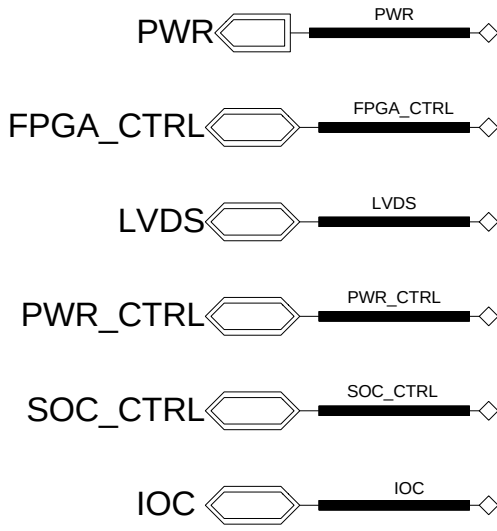
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1) - Route as differential pair signals.
 - Differential impedance: 100ohm (±10%)
 - Maximum intra-pair skew: 0.127mm (5mils)

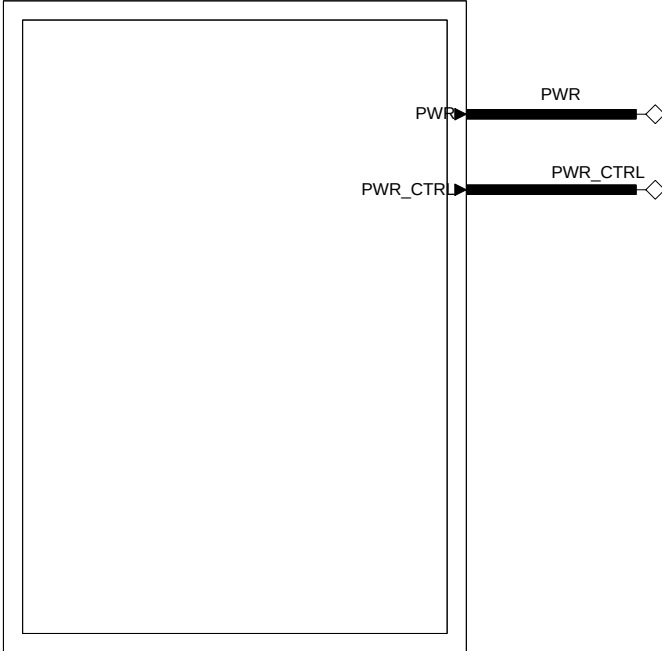
Marvell recommends using Varistors for ESD protection.
 TDK part # AVRH10C101KT1R1NE8

Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

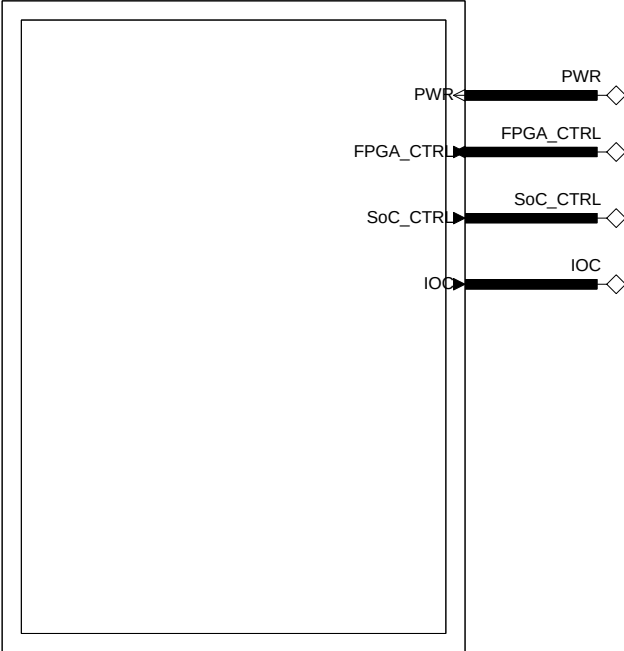
Rev.	type of change		change-no.	date of change	auth. checked
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schematic			Sheet 6	Sheets of 31	Sample
4010_ETH_PHY			Type	Variant	Part Number
substitute for:		substitute by:			



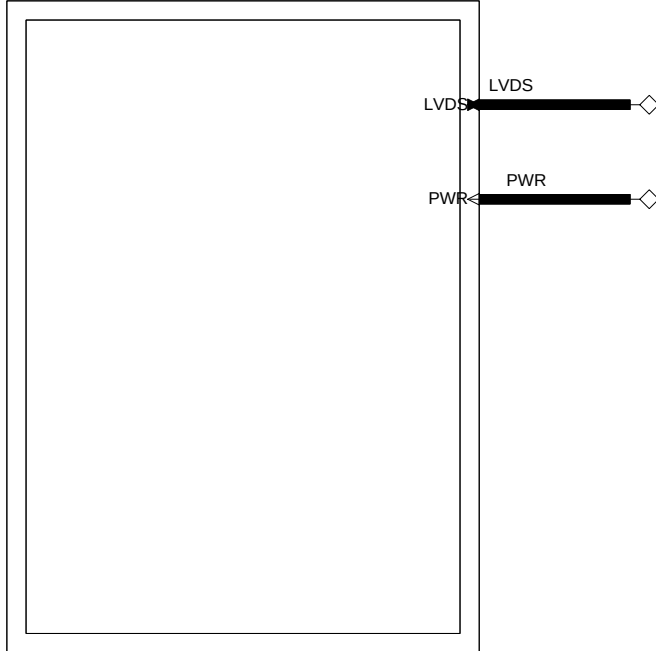
FPGA_POWER



FPGA_CONF

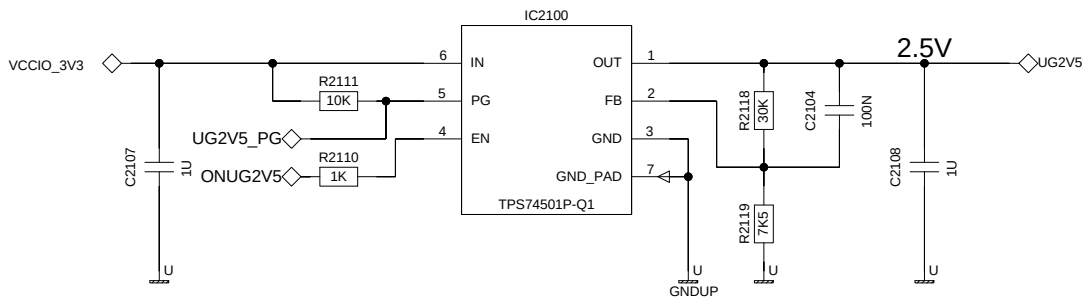
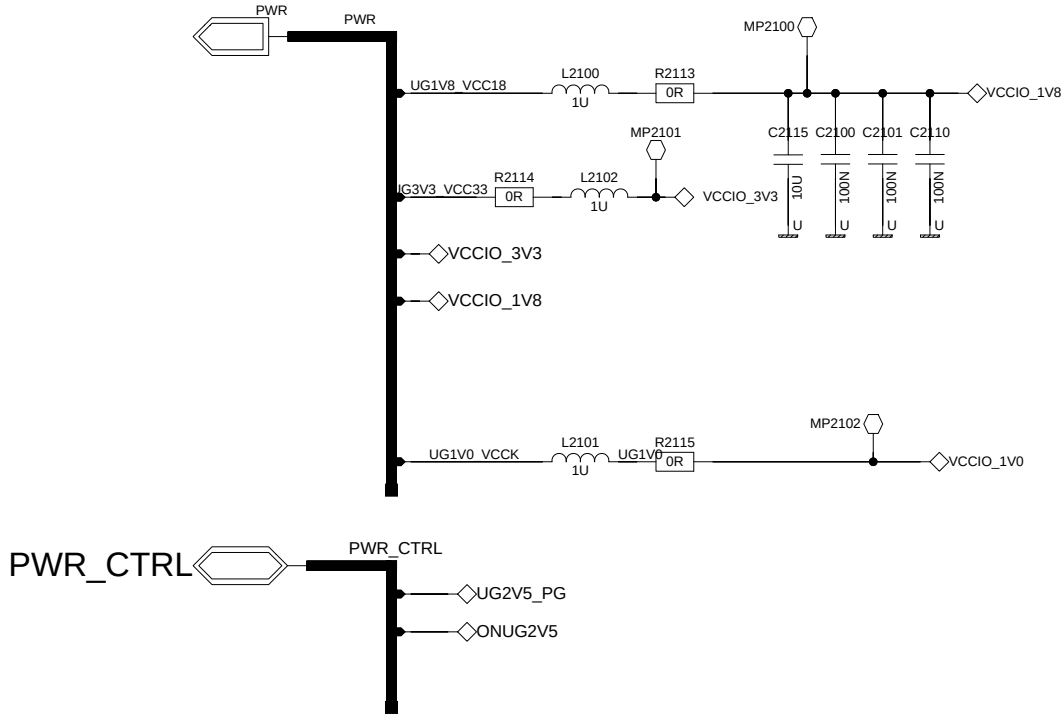
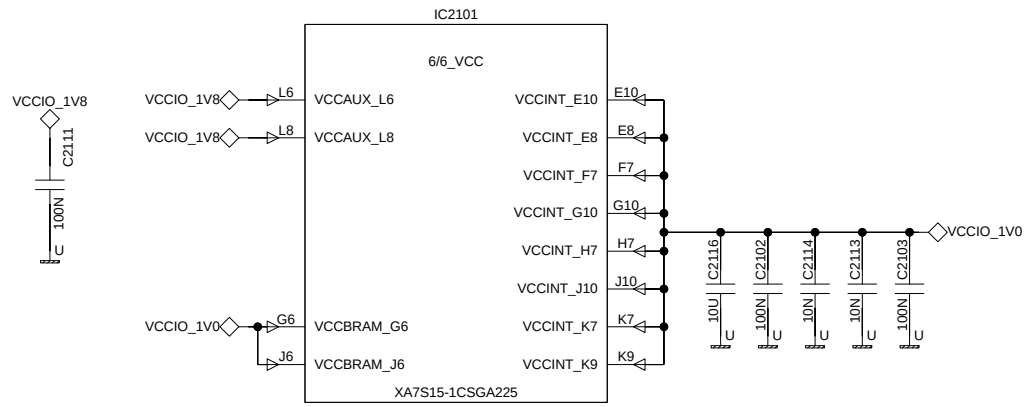
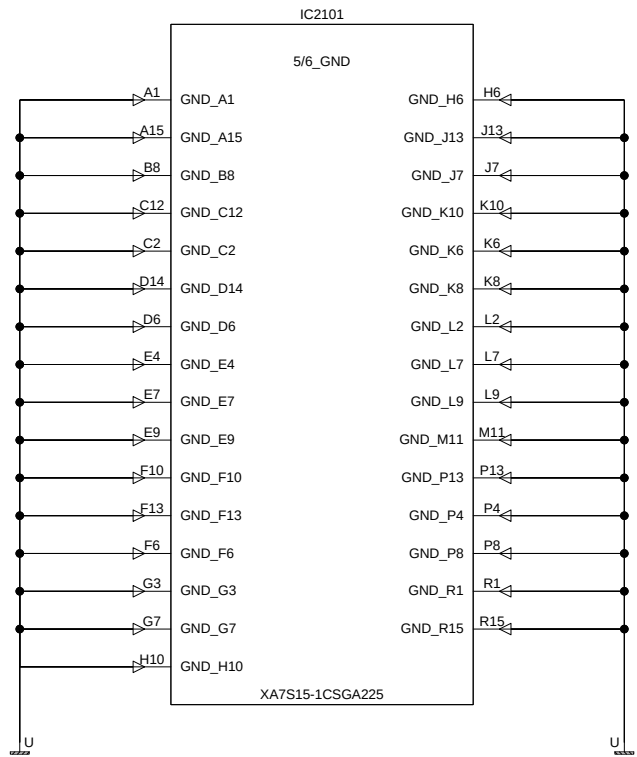


FPGA_VIDEO

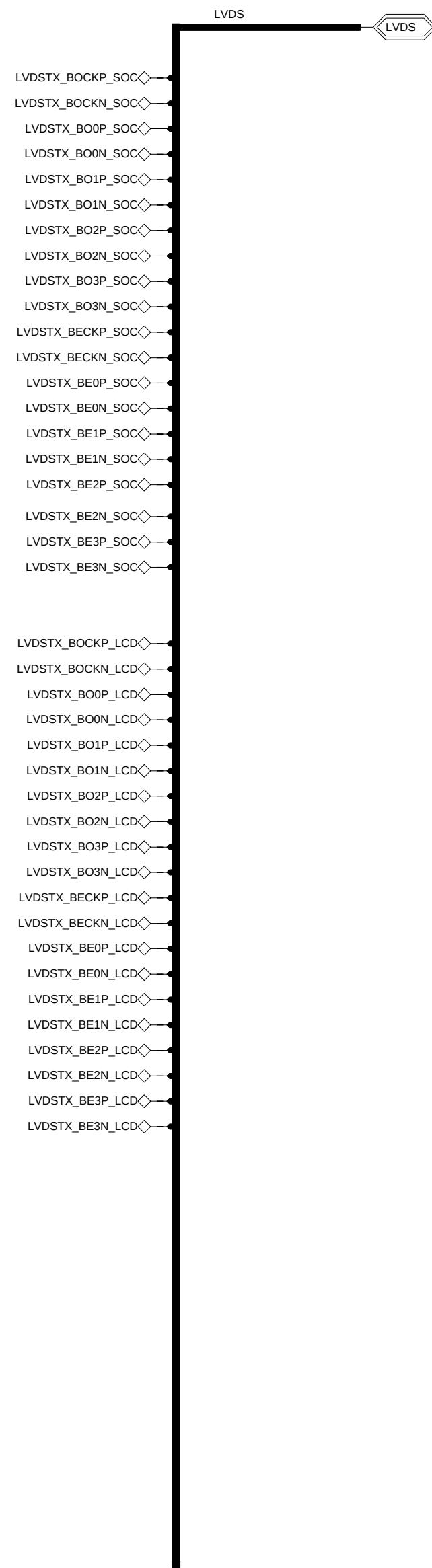
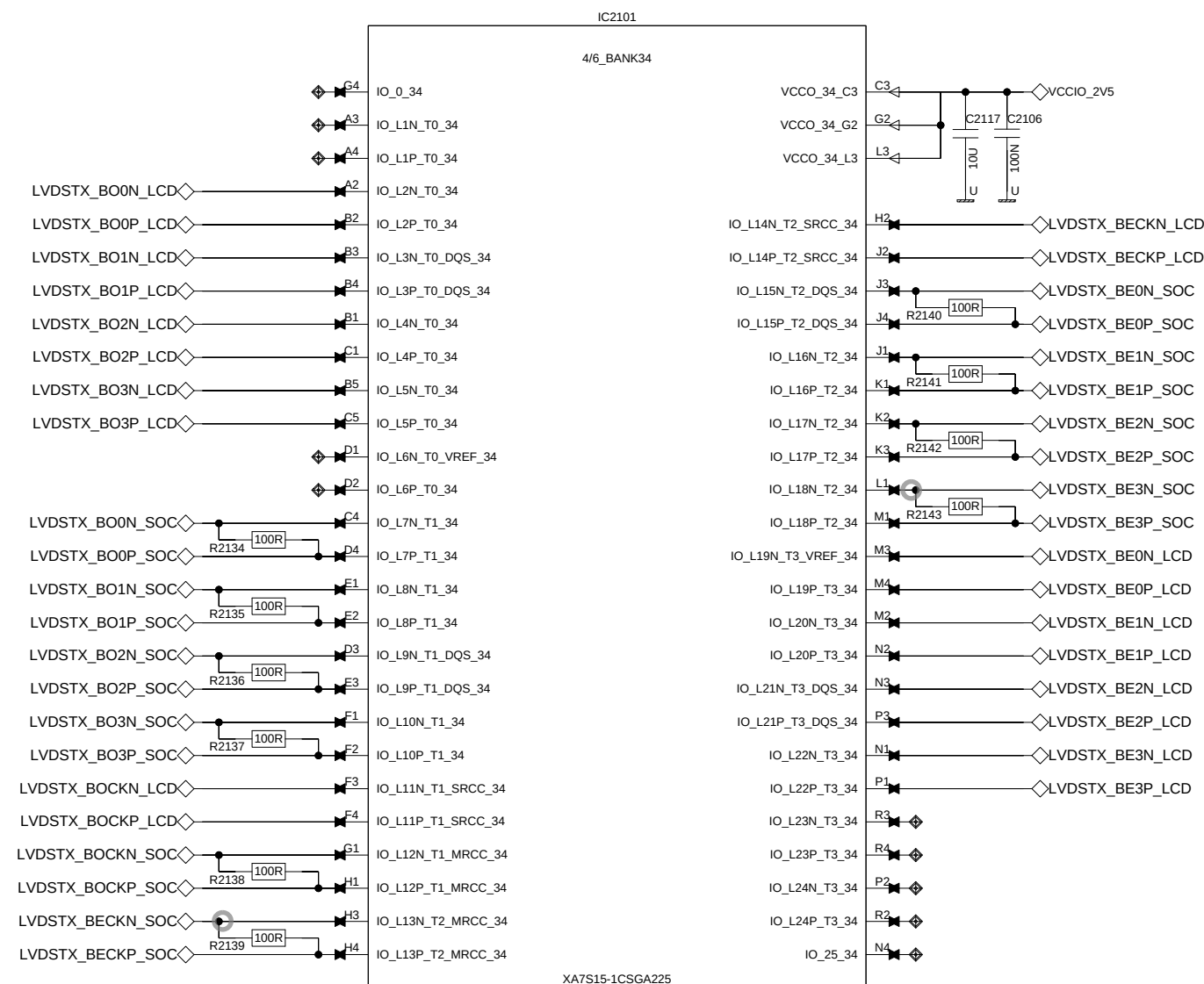
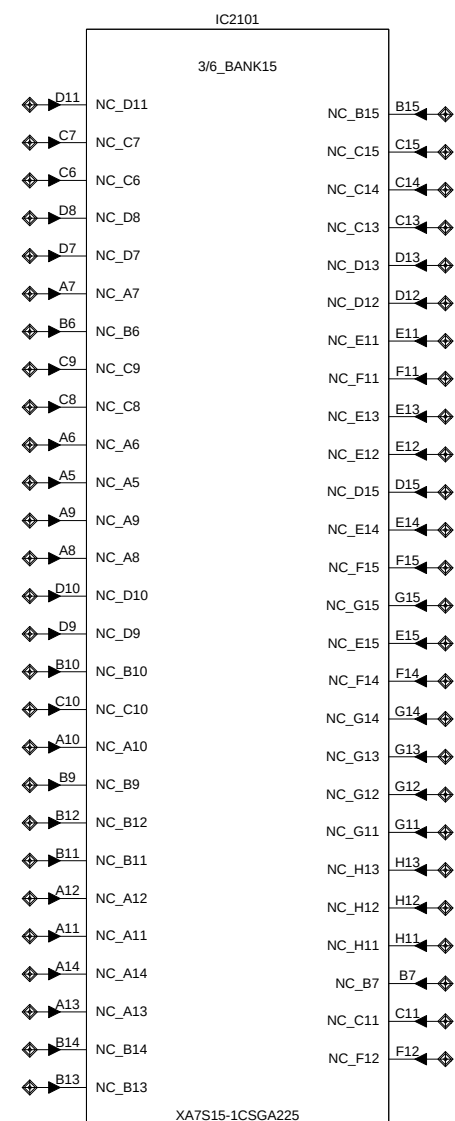
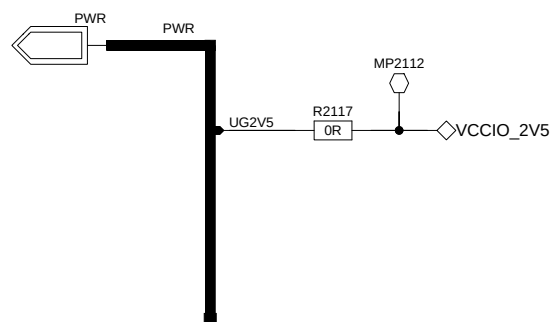


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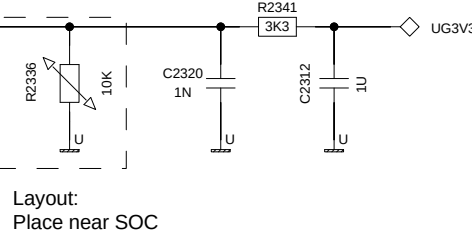
Rev.	type of change		change-no.	date of change	auth. checked		
	Copyright reserved	Writing modifications reserved	date		name		
			created				
			checked				
schematic			Sheet 7	Sheets of 31	Sample		
4030_FPGA			Type	Variant	Part Number		
substitute for:			substitute by:		C143		



Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	date		name	
		created			
		checked			
schematic	Sheet 9		Sheets of 31		Sample
FPGA_POWER	Type		Variant		Part Number
substitute for:		substitute by:		C143	



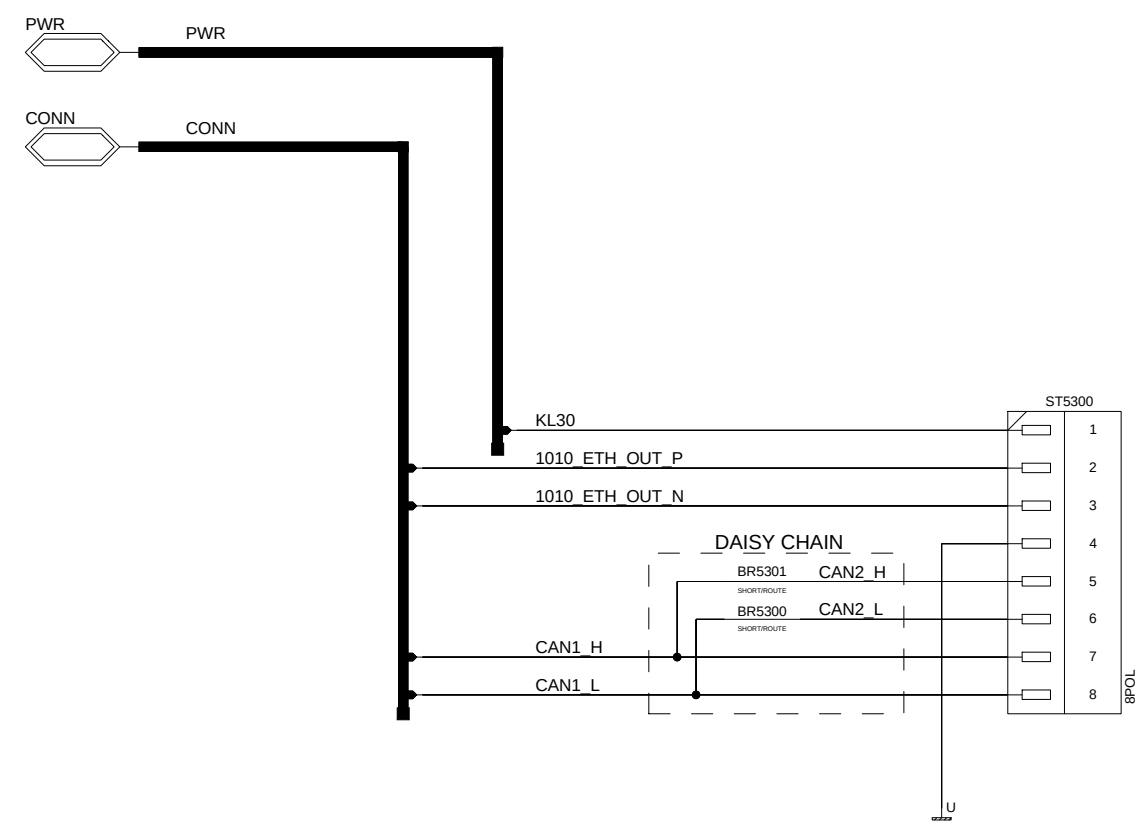
Rev.	type of change		change-no.	date of change	auth. checked
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				created	
				checked	
schematic			Sheet 10	Sheets of 31	Sample
FPGA_VIDEO			Type	Variant	Part Number
substitute for:		substitute by:		C143	



RH550 INTERRUPT TABLE		
INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	
INTP2	ISO	
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	-----
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	-----
INTP14	AWO	DEBUG_RX
INTP15	AWO	

Rev.	type of change					change no.	date of change		auth.	checked
 Copyright reserved Writing modifications reserved						date		name		
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schematic						Sheet 11	Sheets of 31	Sample		
4010_IDC						Type	Variant C143		Part Number	
substitute for:				substitute by:						

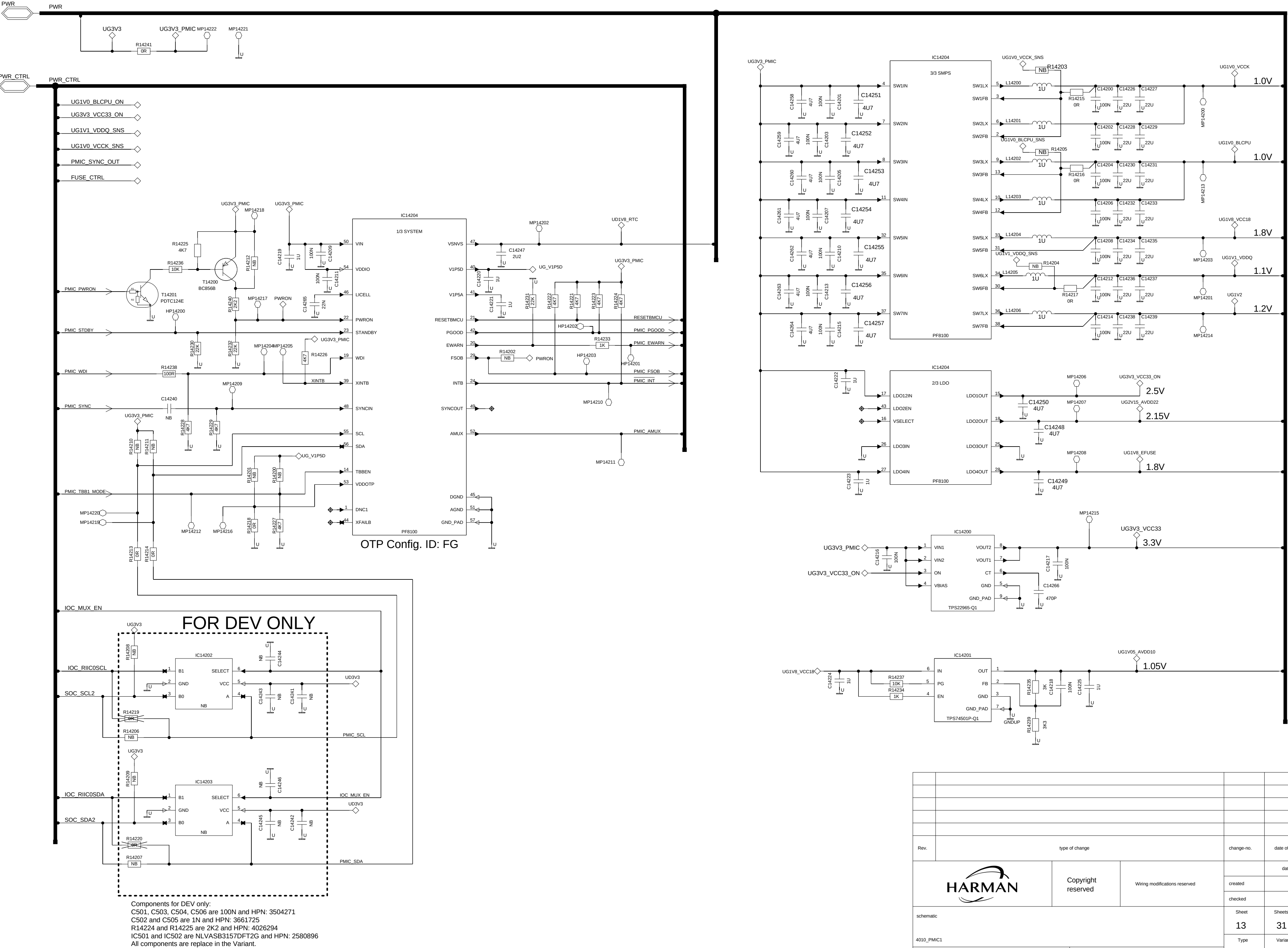
MAIN CONNECTOR (5300-5399)



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		Copyright reserved	Wiring modifications reserved	date		name
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				checked		
schematic			Sheet	Sheets of		Sample
4010_Main_Connector			12	31		
substitute for:			Type	Variant		Part Number
substitute by:						

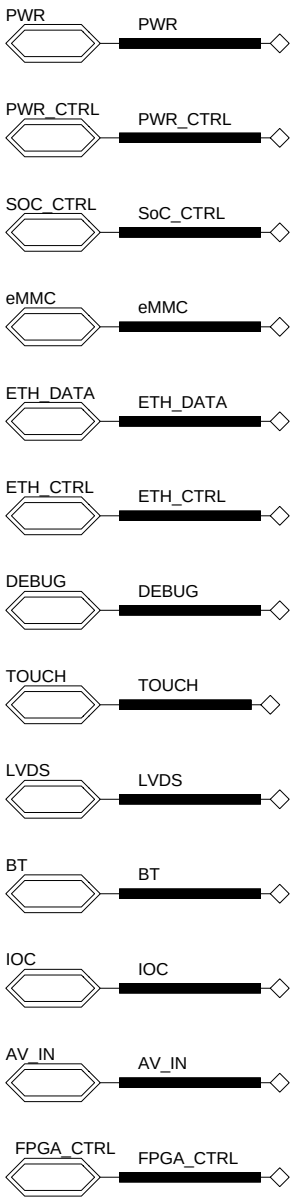
PMIC1 (14200-14299)



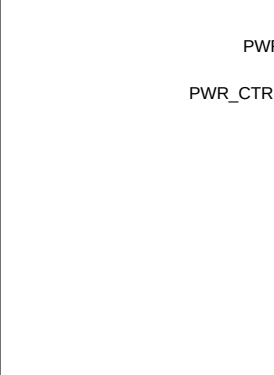
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Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
schematic			Sheet 13	Sheets of 31	Sample
4010_PMIC1			Type	Variant	Part Number
substitute for:		substitute by:			

SOC (10000-10999)

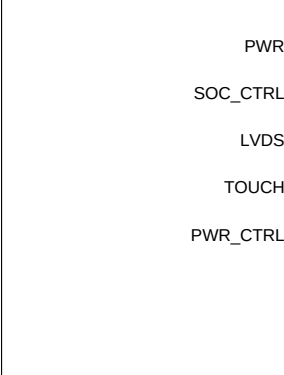


4010 SoC Power



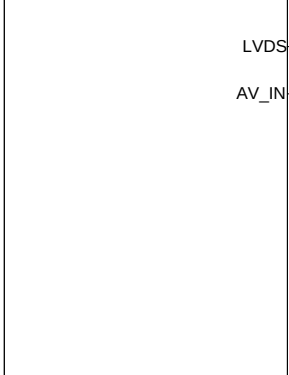
10500 - 10599

4010 SoC_CTRL



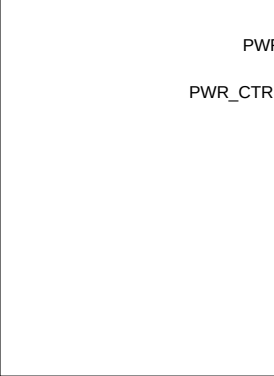
10600 - 10699

4010 SoC Video



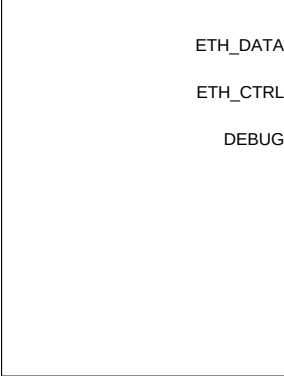
10700 - 10799

4010 SoC LPDDR4



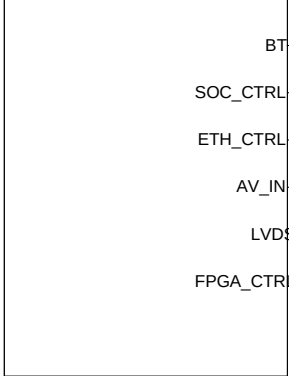
10400 - 10499

4010 SoC_ETH_PCIE_USB



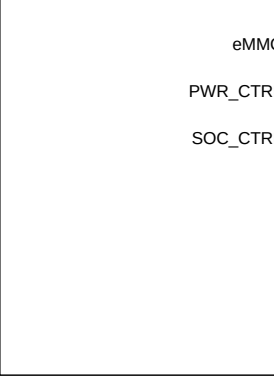
10300 - 10399

4010 SoC Audio



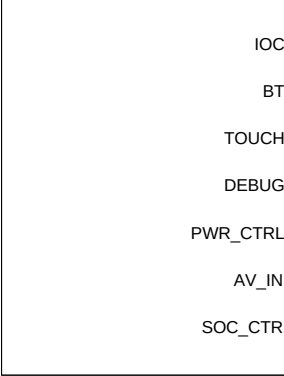
10800 - 10899

4010 SoC_eMMC_NAND_SD



10100 - 10199

4010 SoC_Serial_IF

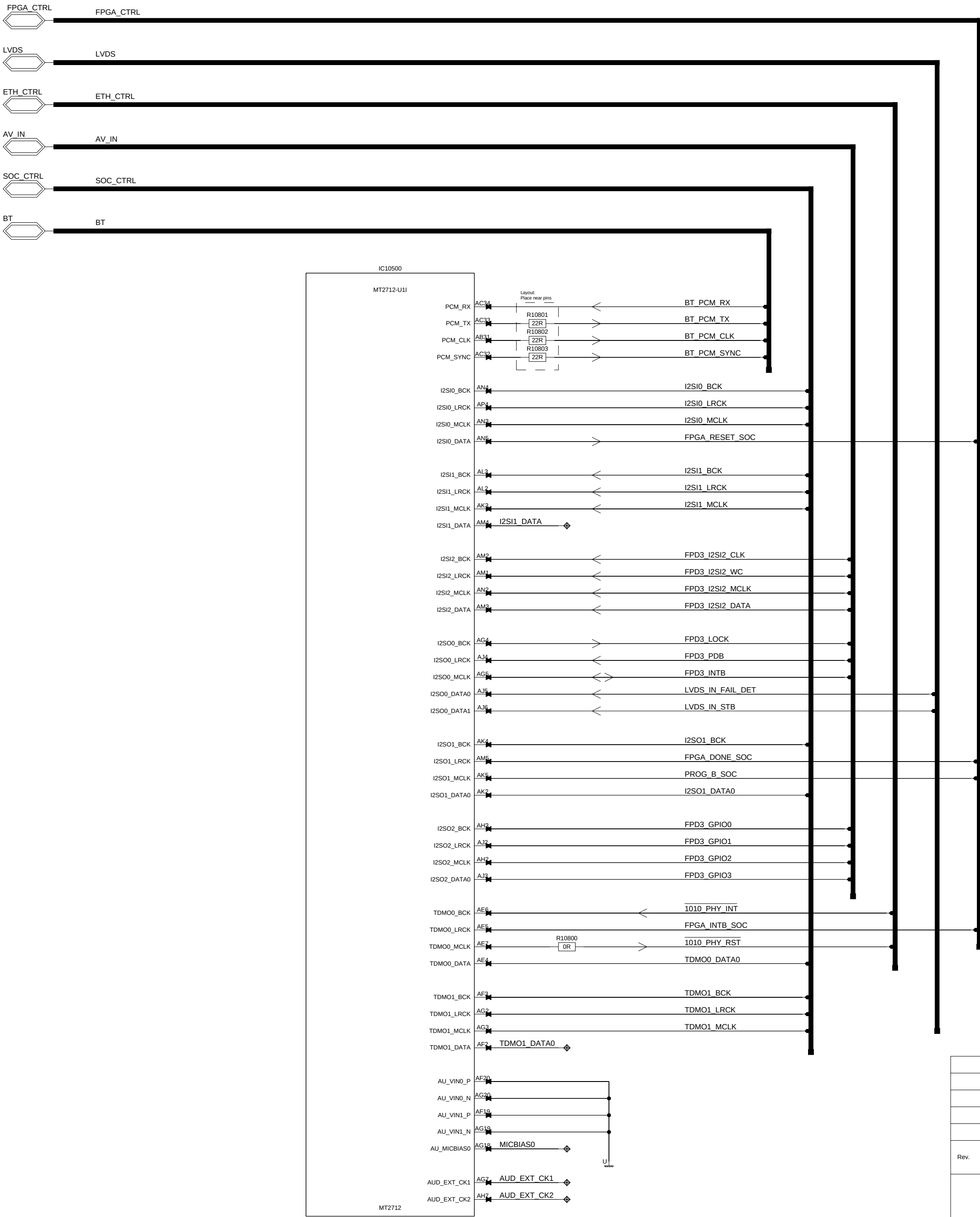


10200 - 10299

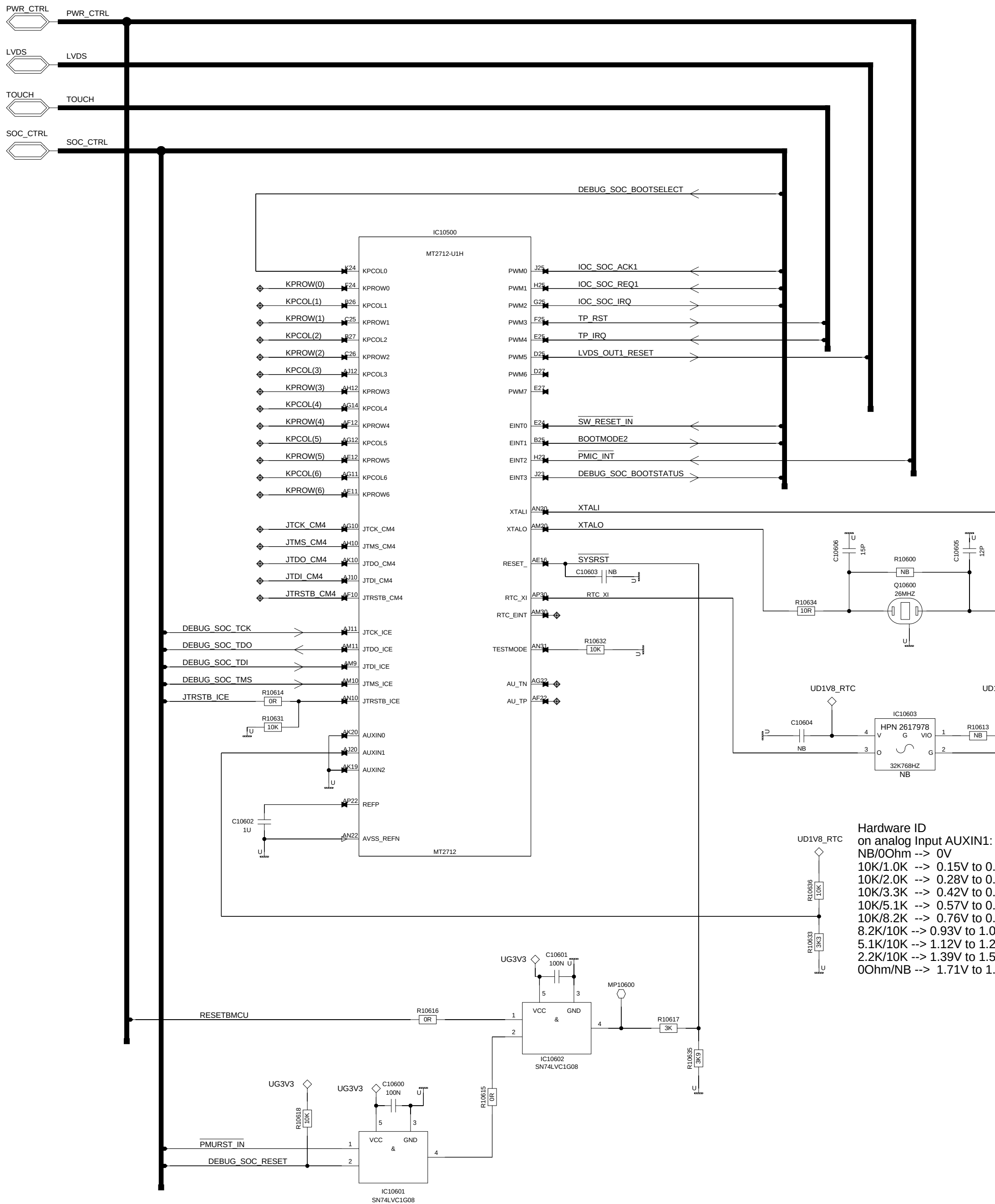
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Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	Wiring modifications reserved	date		name
			created		
			checked		
schematic		Sheet	Sheets of	Sample	
4010_SoC		15	31		
substitute for:		Type	Variant	Part Number	
		substitute by:			

SOC_AUDIO (10800-10899)



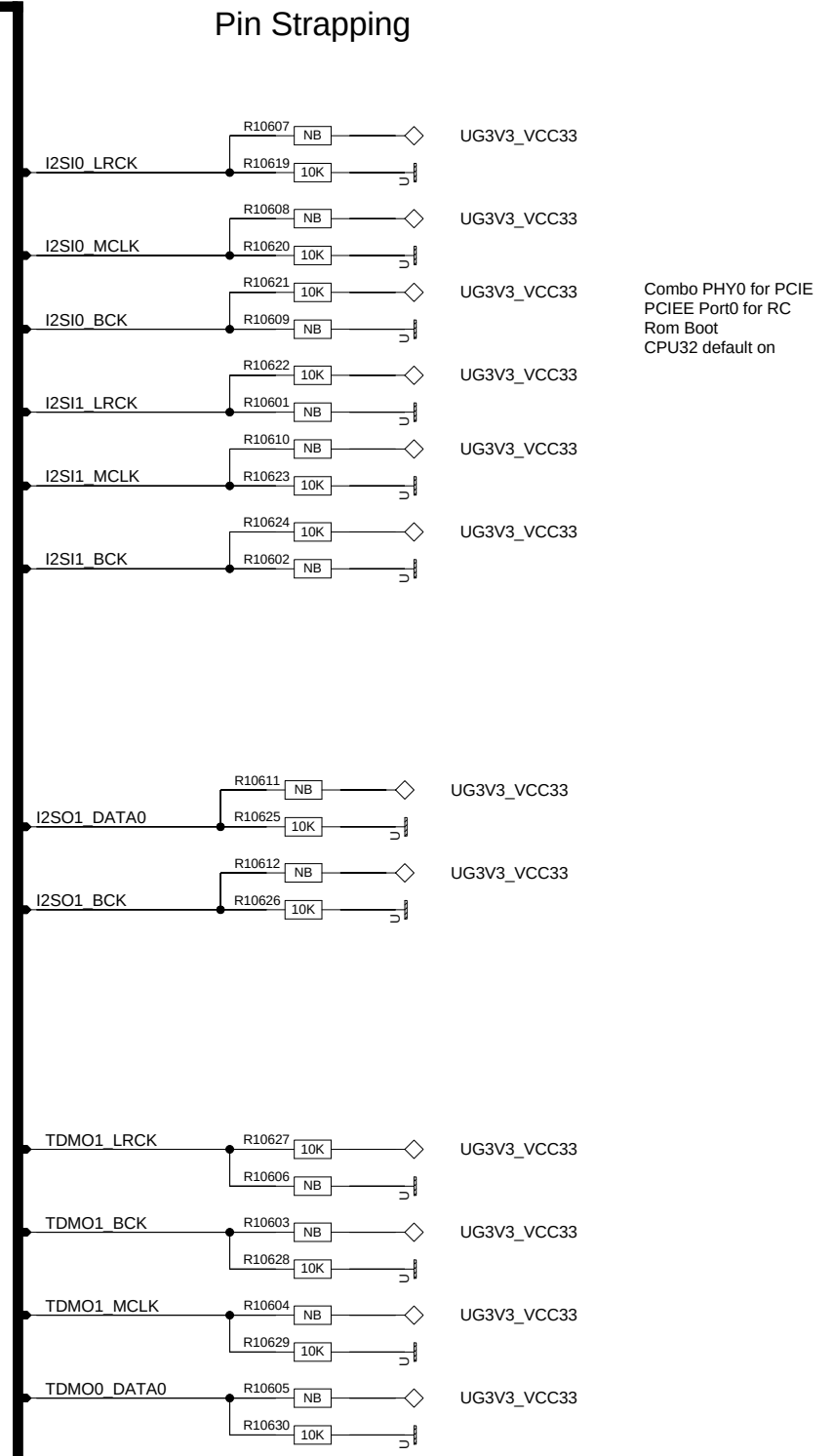
Rev.	type of change	change-no.	date of change	auth.	checked
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			created		
			checked		
schematic		Sheet	Sheets of	Sample	
4010_SoC_Audio		16	31		
substitute for:		Type	Variant	Part Number	
substitute by:					



	multi-used				multi-used	
Mode	I2SiO_LRCK	I2SiO_MCLK	I2SiO_BCK	I2Si1_BCK	I2Si1_LRCLK	I2Si1_MCLK
Combo PHY0 for USB	X	X	X	X	X	1
Combo PHY0 for PCIE	X	X	X	X	X	0
Combo PHY1 for USB	X	X	X	X	1	X
Combo PHY1 for PCIE	X	X	X	X	0	X
PCIEE Port0 for RC	X	X	X	1	X	X
PCIEE Port0 for EP	X	X	X	0	X	X
PCIEE Port1 for RC	X	X	1	X	X	X
PCIEE Port1 for EP	X	X	0	X	X	X
Rom Boot	X	0	X	X	X	X
Nor-flash bypass rom boot	X	1	X	X	X	X
CPU32 default on	0	X	X	X	X	X
CPU64 default on	1	1	X	X	X	X

i2S01_DATA0 status[1]	L	L	H	H
i2S01_BCK status[0]	L	H	L	H
1	EMMC	NAND	NOR	SD
2	SD	EMMC	NAND	NOR
3	NOR	SD	EMMC	NAND
4	NAND	NOR	SD	EMMC

Trapping Mode	TDMO1_LRCK	TDMO1_BCK	TDMO1_MCLK	TDMO1_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1

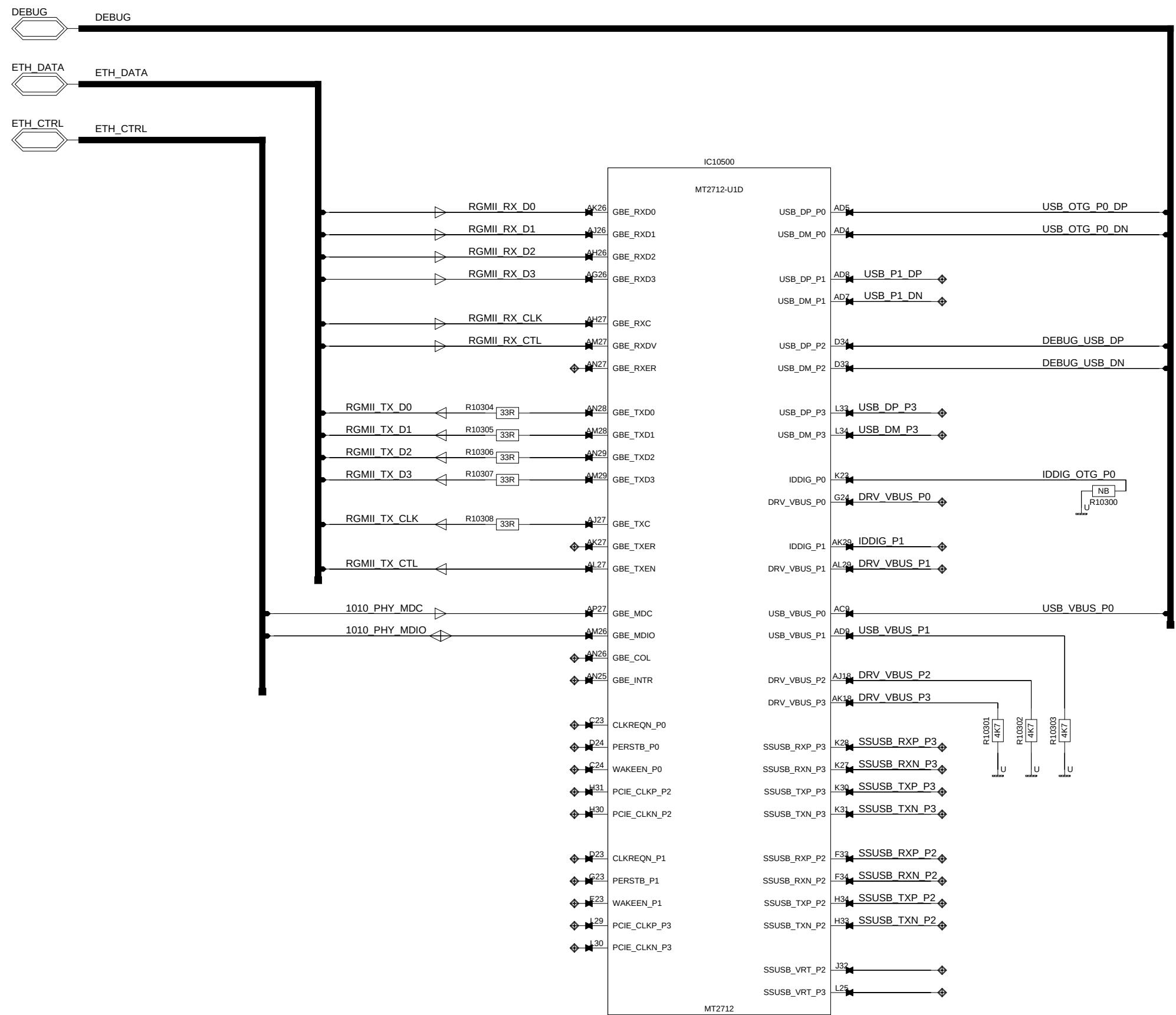


```
Hardware ID
NB/OOHm Input AUXIN1:
10K/1.0K --> 0.15V to 0.18V
10K/2.0K --> 0.28V to 0.32V
10K/3.3K --> 0.42V to 0.47V
10K/5.1K --> 0.57V to 0.65V
10K/8.2K --> 0.76V to 0.86V
8.2K/10K --> 0.93V to 1.05V
5.1K/10K --> 1.12V to 1.26V
2.2K/10K --> 1.39V to 1.55V
OOHm/NB --> 1.71V to 1.89V
```

ED1 HW
ED2 HW
ED3 HW
DV HW
PV HW
PV2 HW
PV3 HW

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 17	Sheets of	Sample
4010_SoC_CTRL			Type	Variant	Part Number
substitute for:		substitute by:			

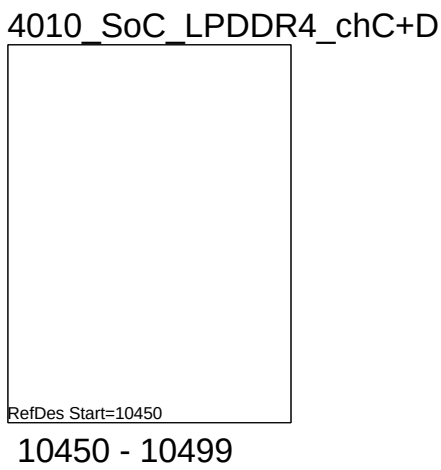
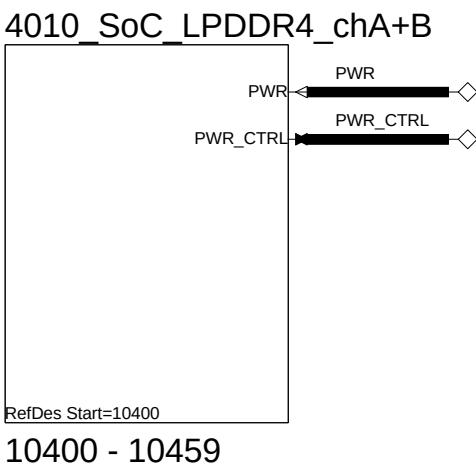
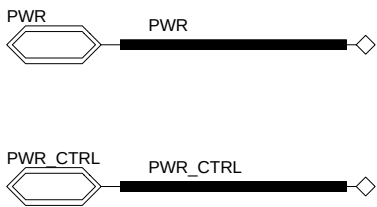
SOC_ETH_PCl_e_USB (10300-10399)



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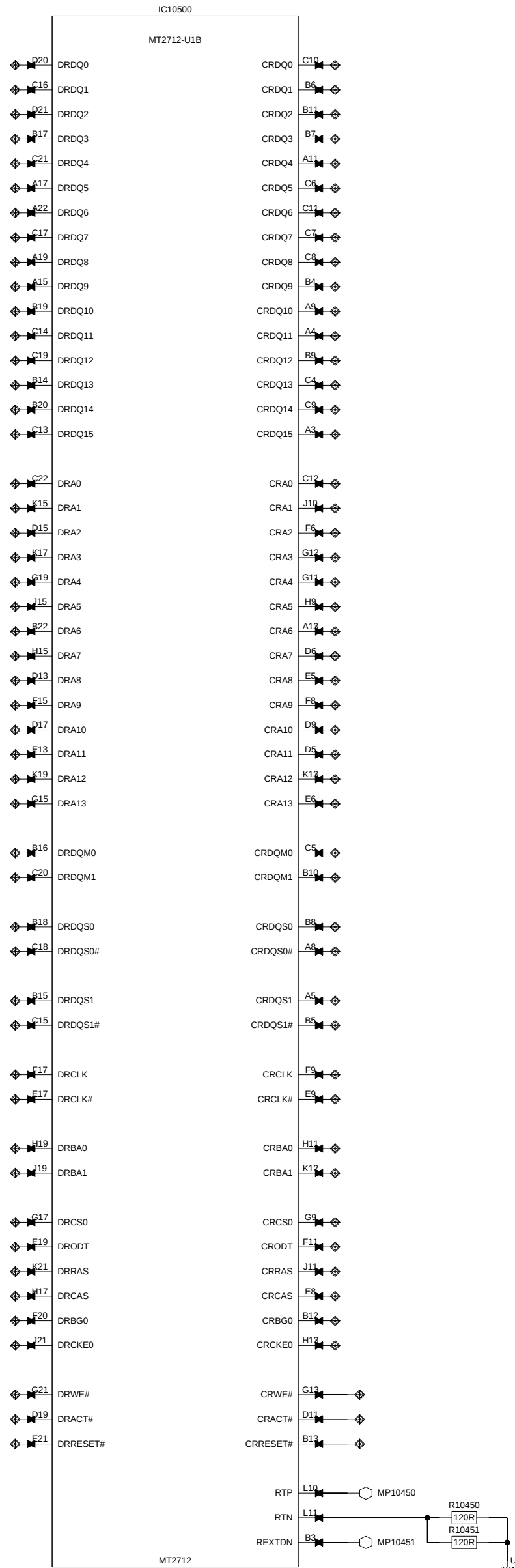
Rev.	type of change		change-no.	date of change	auth. checked	
		Copyright reserved	Wiring modifications reserved	date name		
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				checked		
schematic			Sheet	Sheets of	Sample	
4010_SoC_ETH_PCl_e_USB			18	31		
substitute for:		substitute by:		Type	Variant	Part Number

SOC_LPDDR4 (10400-10499)



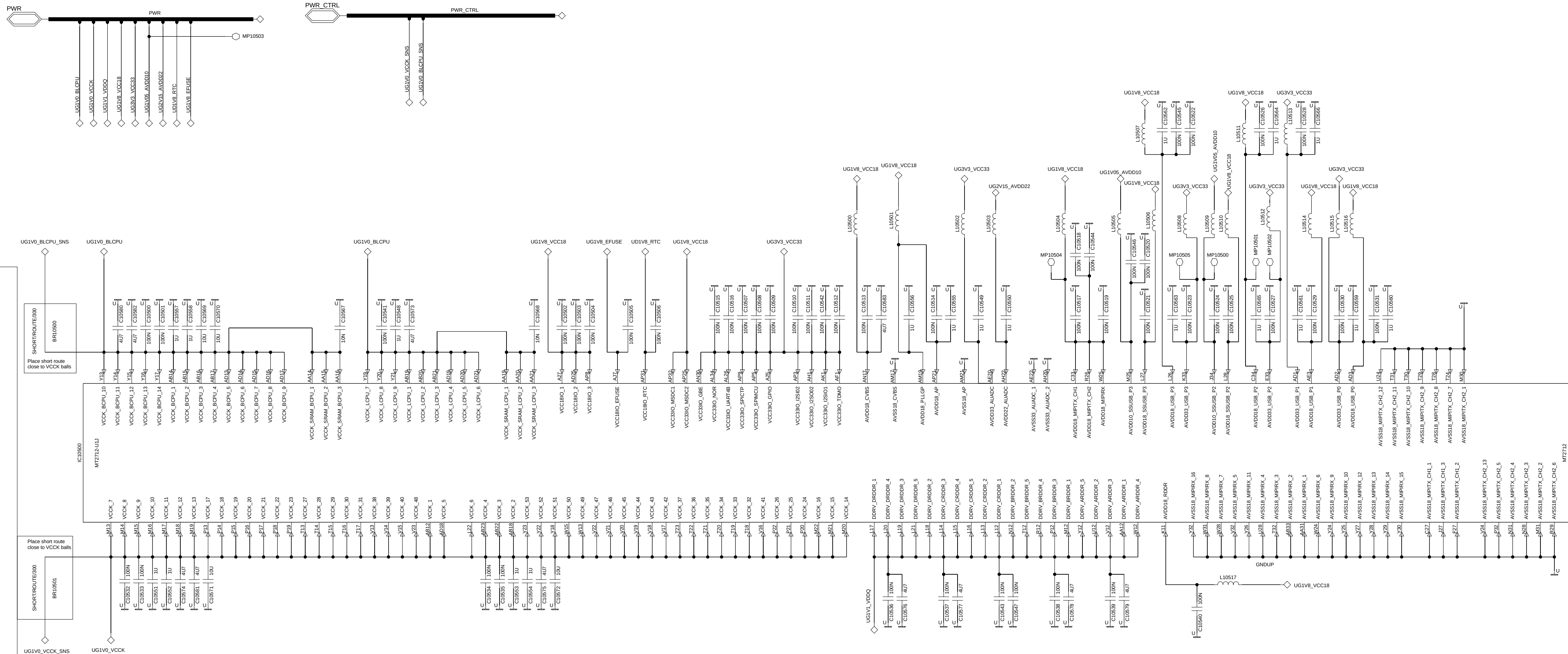
Rev.	type of change		change-no.	date of change	auth.	checked
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				checked		
schematic			Sheet	Sheets of	Sample	
4010_SoC_LPDDR4			19	31		
substitute for:			Type	Variant	Part Number	
substitute by:						

SOC_LPDDR4 (10450-10499)

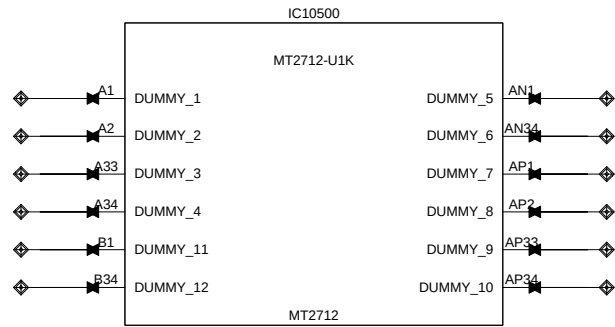


Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_SoC_LPDDR4_chC+D			21	31	
substitute for:		substitute by:	Type	Variant	Part Number

SOC_POWER (10500-10599)



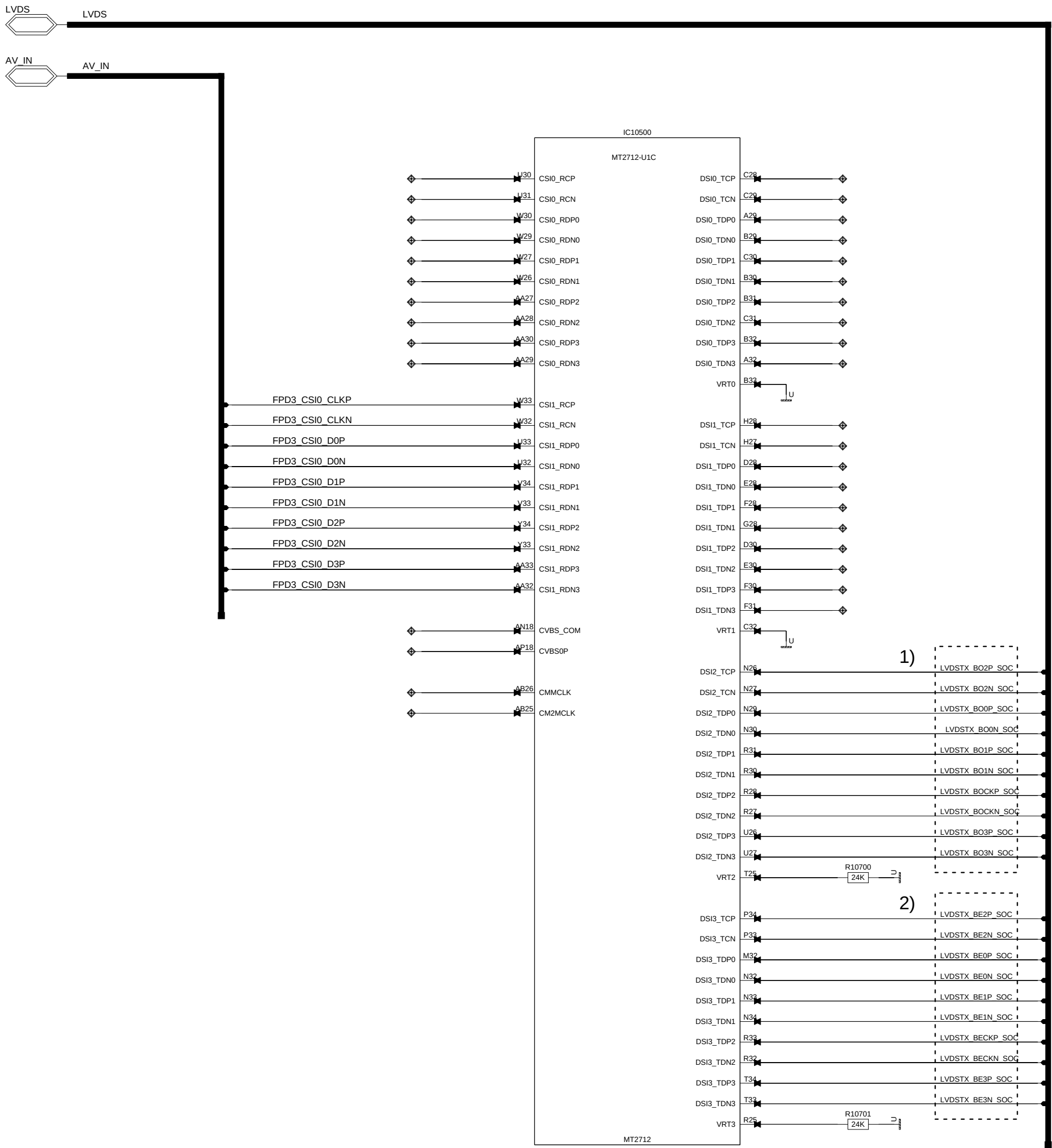
4010 SoC GND



Pin connection diagram for the MT2712-U1E IC. The diagram shows various external components connected to the IC pins. On the left, components include SOC_CTRL, IOC, BT, DEBUG, AV_IN, PWR_CTRL, and TOUCH. On the right, components include TP_SDA, TP_SCL, SOC_SDA2, SOC_SCL2, FPD3 I2C5 SDA, FPD3 I2C5 SCL, DEBUG SOC UART RXD, DEBUG SOC UART TXD, BT URXD3, BT UTXD3, BT URTS3, BT UCTS3, SOC_FPGA MISO EEPROM, SOC_FPGA MOSI EEPROM, SOC_FPGA SCK EEPROM, SOC_FPGA CSB EEPROM, IOC SPI4 MI, IOC SPI4 MO, IOC SPI4 CK, and IOC SPI4 CS. The IC pins are labeled with names like I2C_SDA0, I2C_SCL0, I2C_SDA1, I2C_SCL1, I2C_SDA2, I2C_SCL2, I2C_SDA3, I2C_SCL3, I2C_SDA4, I2C_SCL4, I2C_SDA5, I2C_SCL5, URXD0, UTXD0, URXD1, UTXD1, URXD2, UTXD2, URXD3, UTXD3, URTS3, UCTS3, URXD4, UTXD4, URTS4, UCTS4, URXD5, UTXD5, URTS5, UCTS5, SPI0_MI, SPI0_MO, SPI0_CK, SPI0_CSN, SPI1_SI, SPI1_SO, SPI1_CK, SPI1_CSN, SPI2_MI, SPI2_MO, SPI2_CK, SPI2_CSN, SPI3_MI, SPI3_MO, SPI3_CK, SPI3_CSN, SPI4_MI, SPI4_MO, SPI4_CK, SPI4_CSN, SPI5_MI, SPI5_MO, SPI5_CK, SPI5_CSN. The IC is labeled MT2712-U1E.

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 24	Sheets of 31	Sample
4010_SoC_Serial_IF			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_VIDEO (10700-10799)



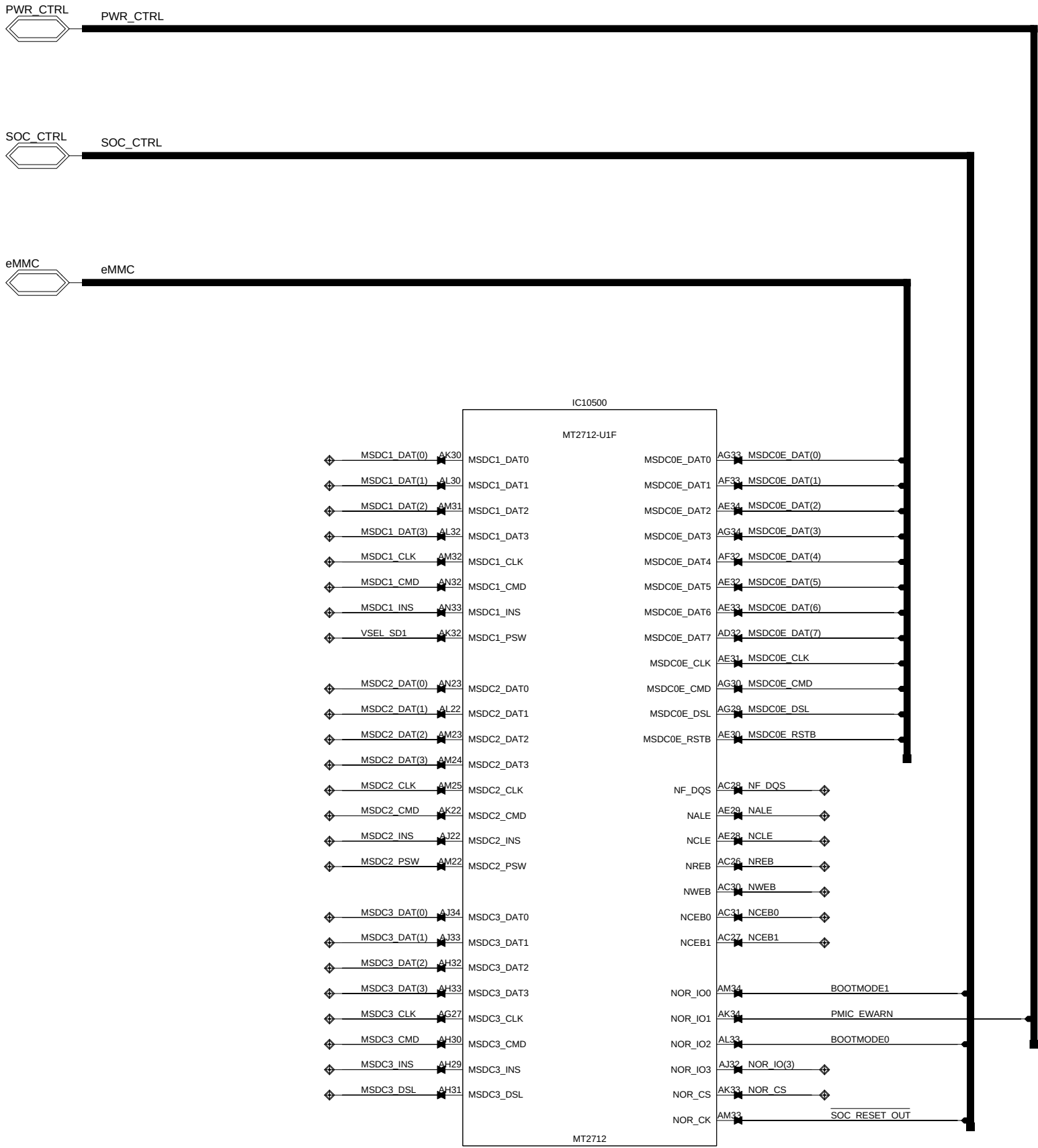
- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BOCK) to data (BO[0-3]): 5mm

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of	Sample	
4010_Soc_Video			25	31		
substitute for:			Type	Variant	Part Number	
substitute by:						

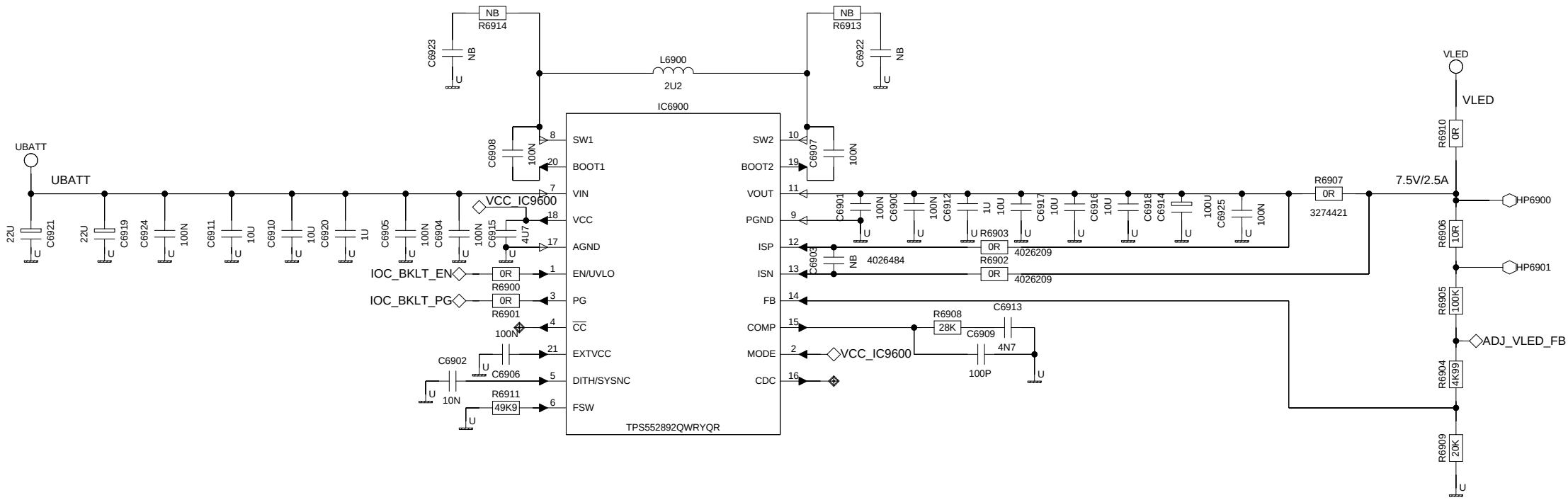
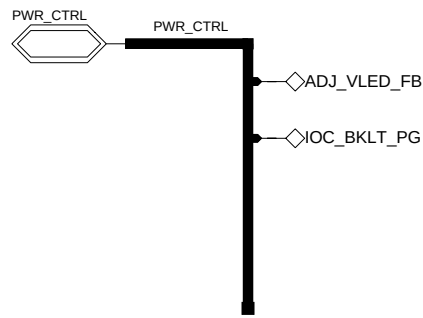
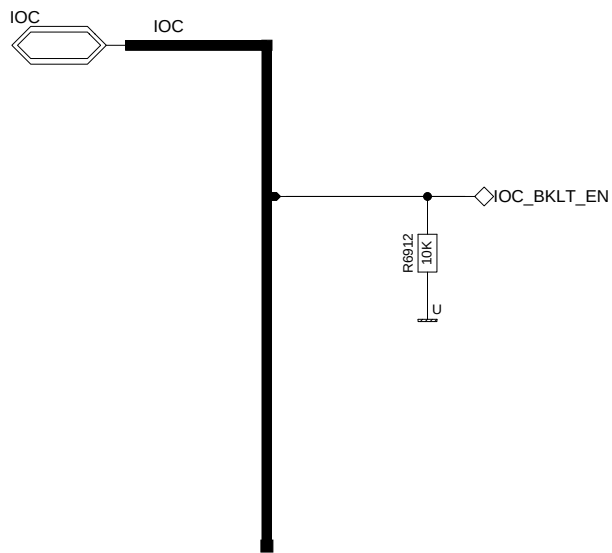
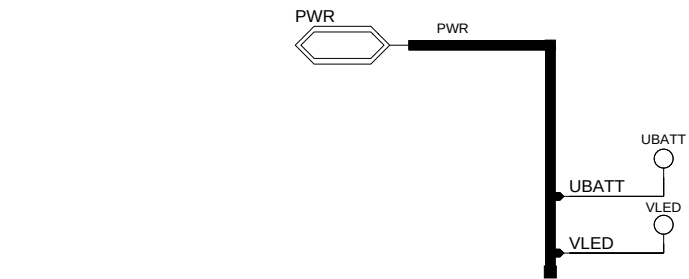
A2 00 02 12

SOC_eMMC_NAND_SID (10100-10199)



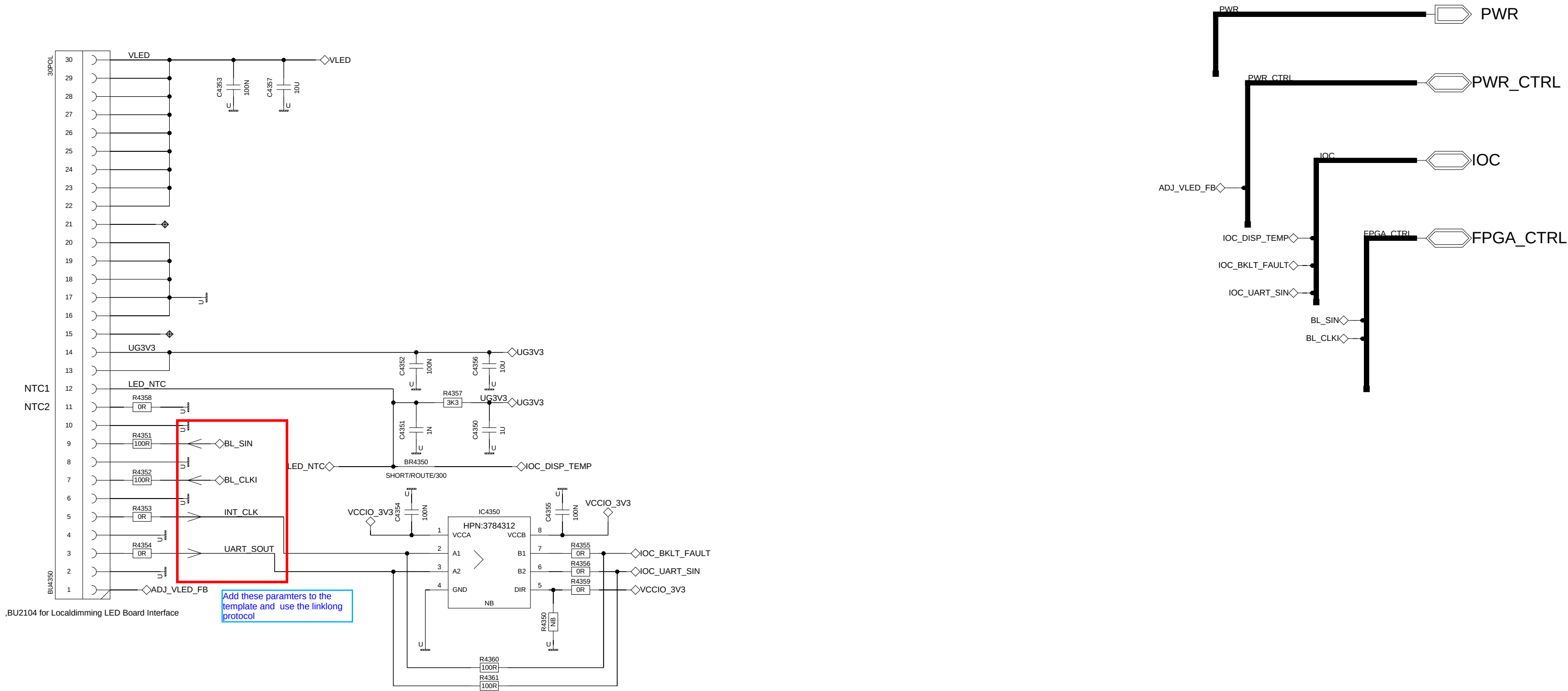
Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet 26	Sheets of 31	Sample
4010_SoC_eMMC_NAND_SD				Type	Variant	Part Number
substitute for:		substitute by:				

TFT BL (6900-6999)

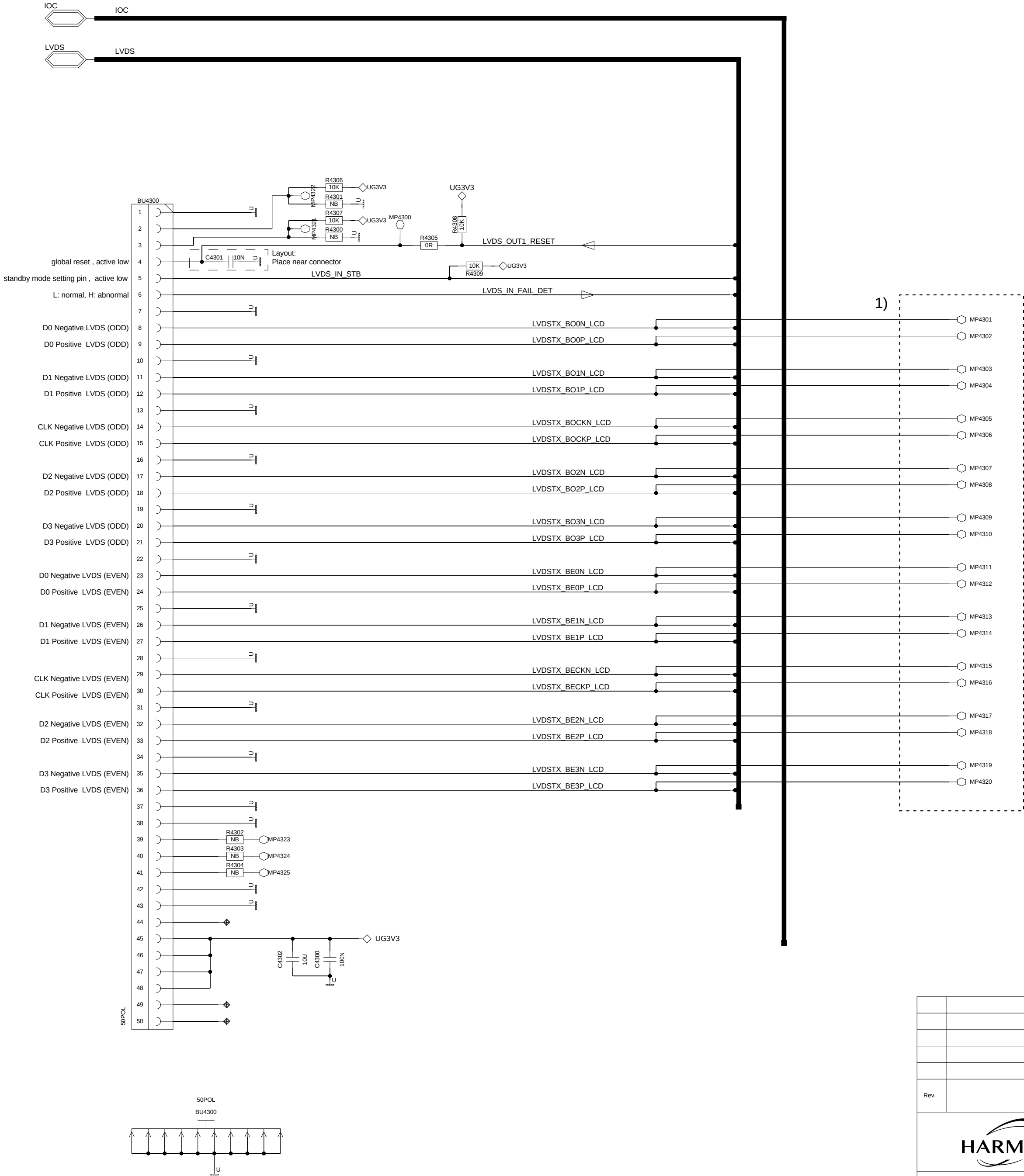
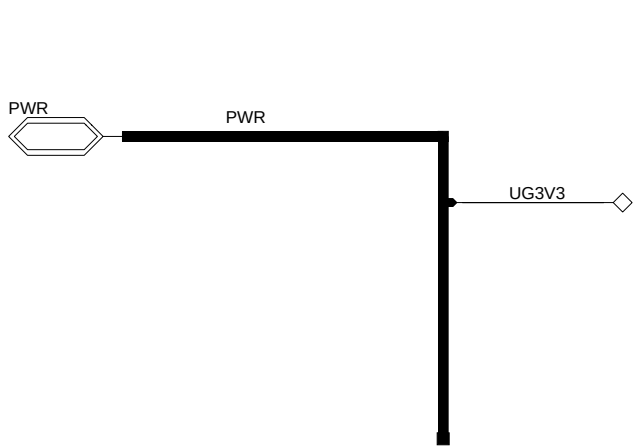


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			date	name	
		created			
		checked			
schematic		Sheet	Sheets of	Sample	
4010_TFT_BL		Type	Variant	Part Number	
substitute for:	substitute by:		C143		

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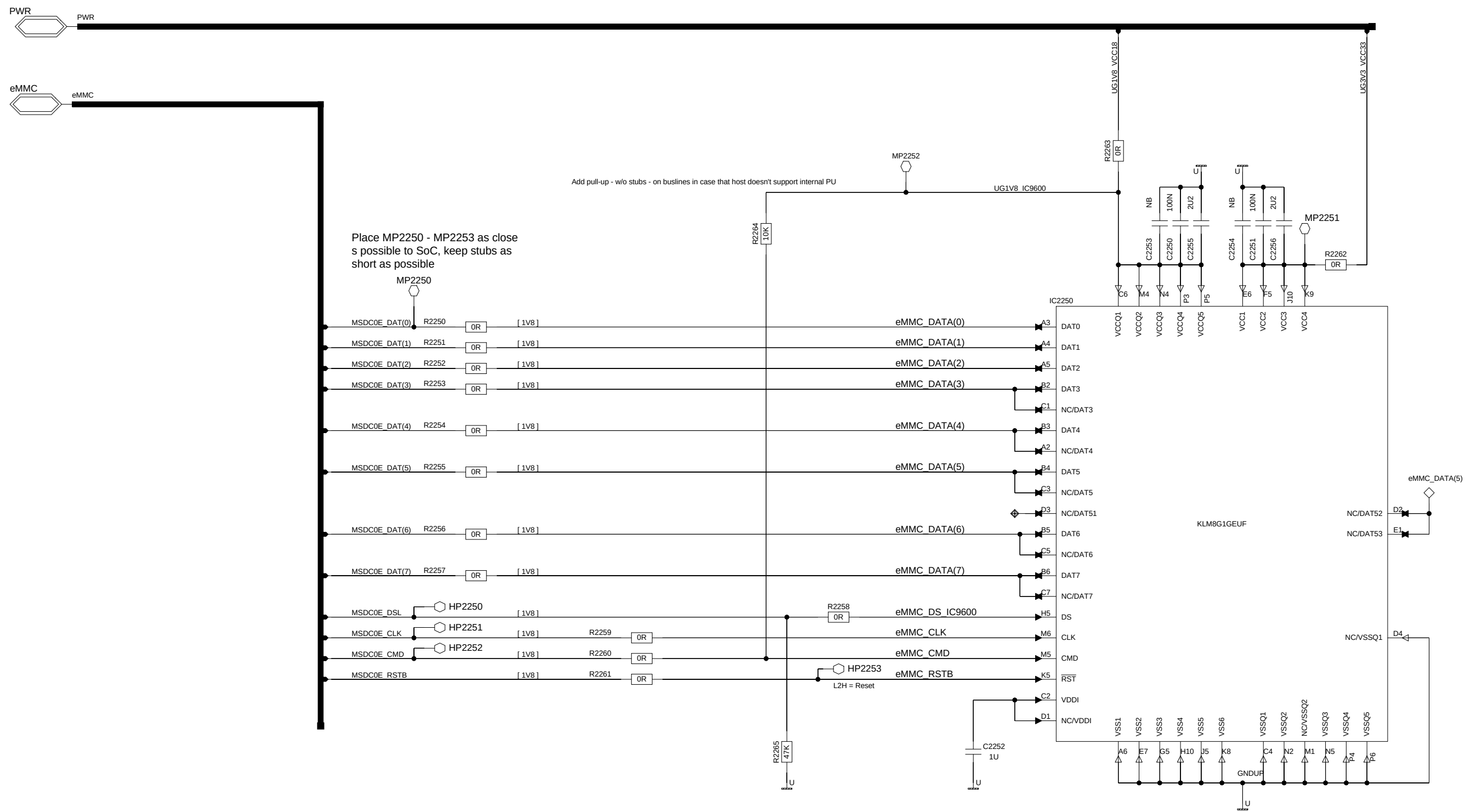
Rev.	type of change		change-no.	date of change	auth. checked
 Copyright reserved Wiring modifications reserved			date		name
			created		
			checked		
schematic			Sheet	Sheets of	Sample
4010_TFT_BL_Conn			28	31	
substitute for:			Type	Variant	Part Number
substitute by:			C143		



Verify pin out during layout

1) Testpoints should be placed as close as possible to an existing VIA

eMMC (2250-2299)



eMMC	Density	Part Number
Micron (-40 to 105°C)	8 GB	3881466
Micron (-40 to 85°C)	8 GB	3543617